

Highlights

Two bases for a voxel, the medium and the wavefield: the error of elastic volume-integral scattering, isolated on a space-filling layer of cubes

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- A voxel needs two bases: one for the medium and one for the wavefield
- On a layer of cubes the discretisation error is isolated, with no shape error
- The order is $\min(2p+2, 2r+2)$ for field degree p and medium degree r
- The single-site T-matrix's error is the medium's relative projection error
- The error is known in advance, from the medium and the grid alone

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Abstract

To model seismic waves efficiently in a realistically complicated medium, volume-integral methods divide it into cubic voxels coupled through the background Green's tensor. Each voxel must represent the medium within it and the wavefield within it. We ask which basis each needs and what error a given choice leaves. We answer on a layer of voxels: because cubes tile space, n planes of voxels are the layer itself, and its exact solution isolates the discretisation error. With the wavefield expanded to Legendre degree p and the medium's contrast to degree r in each voxel, the scheme converges at order $\min(2p + 2, 2r + 2)$, with leading error $c_p(kd)^{2p+2}$ in closed form; the orders hold for SV and SH waves and at oblique incidence. The single-site T -matrix's second-order error equals the relative projection error of the medium at degree $\min(p, r)$: the two bases must be matched, and the error is known from the medium before anything is solved. In a plane-stratified background the orders persist: the background changes the field the voxels must represent, not how well they represent it. The single site is the first member of a hierarchy in the gradients of the field, whose coefficients are moments of the

Green’s tensor over the cube, in closed form on four constants. Its blocks gain accuracy in pairs: with third gradients it is three hundred times more accurate than the uniform-strain closure against the exact sphere, and as a fourth-order voxel scheme it matches the first-moment voxel on a smoothly graded sphere.

Keywords: Elastic wave scattering, Volume integral equation, Foldy–Lax multiple scattering, Discrete dipole approximation, Galerkin method, Convergence analysis

1. Introduction

The problem this paper serves. Seismic waves in the crust are scattered by heterogeneity on every scale, and the aim of the work this paper begins is an efficient model of the wavefield in a medium of realistic complication. The models that describe such a medium are parameterised in many ways (as layers or blocks, by smooth basis functions, or by interfaces), and a calculation that uses one must first sample it onto a grid. A volume-integral scattering calculation divides the medium into cells and treats every cell as a scatterer: a voxel carrying a single-site T -matrix, the operator that turns the field incident on the voxel into the sources it radiates, and coupled to every other voxel through the Green’s tensor of the background, with the multiple-scattering series summed.

The question: two bases. A voxel is a small region in which two functions must be represented by a finite number of coefficients. One is the *medium*: the contrast of the elastic moduli and the density from the background, which the model prescribes. The other is the *wavefield*: the displacement and

17 strain inside the voxel, which the calculation must find. Each is expanded
 18 in a basis on the cell, and the two bases are separate choices. Here both
 19 are Legendre polynomials, the wavefield to degree p and the contrast to
 20 degree r ; the usual scheme, a uniform field in a cell of constant properties,
 21 is $p = r = 0$. Neither choice is new in itself. In higher-order volume integral
 22 methods for electromagnetics the material within an element is represented
 23 by polynomials whose orders are independent of those of the field [1], as it
 24 is in finite elements, and voxels that carry a linear field are in use [2]. What
 25 has been missing is the law that relates the two bases: the order a given pair
 26 attains, and the error it leaves. This paper asks which basis each of the two
 27 needs, and what error a given pair (p, r) leaves. The answer decides the cost
 28 of everything built on it, because a voxel that represents its contents more
 29 accurately can be larger, and the number of voxels, not the size of each one's
 30 state, sets the cost of the multiple-scattering solve.

31 *Why the question is hard to answer.* For the media such a scheme is built for,
 32 there is no exact solution. Its error cannot be measured against the answer.
 33 Comparison with another numerical method mixes the errors of both. And a
 34 refinement study can mislead, because a voxel scheme can converge smoothly
 35 to the wrong limit. §4 shows how that happens with the most natural choice
 36 of coupling, the Green's tensor evaluated between voxel centres: it carries
 37 a local term whose size per unit volume does not shrink as the voxels do,
 38 and unless the single voxel's response is matched to cancel it, refinement
 39 converges to the wrong answer without revealing it.

40 *The method: a layer of voxels.* This paper therefore answers the question
 41 in the simplest configuration in which a voxel scheme runs in full (every

voxel, every coupling, the whole multiple-scattering series, at normal and
 at oblique incidence) and the exact answer is nevertheless known: a layer
 filled with voxels. Because cubes tile space, a layer of identical voxels does
 not approximate a homogeneous layer; it *is* one. Any difference between
 its reflection and transmission and those of the continuous layer is therefore
 error of the discretisation alone, with no modelling error and no shape error
 behind which to hide. A sphere, the usual test object for volume-integral and
 discrete-dipole methods [3], cannot give this: its error mixes the discretisation
 with the staircase approximation of a curved surface by cubes. The layer
 plays the part of the patch test in finite elements: reproducing a uniform
 medium correctly is necessary, though not sufficient, for a scheme to be
 trusted in a heterogeneous one. The experiment is then made harder in
 three steps, each with an exact solution of its own: a randomly stratified
 layer, in which every cell is homogeneous but the cells differ; a layer whose
 properties vary smoothly within every cell, which is where the basis for the
 medium is tested; and a sphere whose contrast falls smoothly to zero at its
 surface, which carries the layer's prediction into three dimensions without a
 staircase. What the layer establishes about the scheme's two ingredients, the
 coupling between voxels and the response of a single voxel, are properties of
 the ingredients, which are the same in any medium. A laterally heterogeneous
 medium remains to be tested.

The answer. (1) *The order is set by the weaker basis.* The scheme converges
 at order $\min(2p + 2, 2r + 2)$ in the cell size (§9). A contrast held constant in
 each cell therefore caps every voxel at second order in a medium that varies
 within the cells, however rich the basis of its field; and a contrast of higher

67 degree than the field is wasted. (2) *The leading error is in closed form, for*
 68 *every grid.* For a voxel of side d and a wave of wavenumber k it is $c_p(kd)^{2p+2}$
 69 with $c_0 = 1/12$, $c_1 = 1/720$ and $c_2 = 1/100800$ (§8); the orders 2, 4 and 6 of
 70 $p = 0, 1$ and 2 hold for P, SV and SH waves at normal and oblique incidence.
 71 For the collocation scheme, the uniform-field voxel in its simplest form, the
 72 error is $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission (§7), and the
 73 opposite signs explain why no correction of the single site can improve both
 74 (§12). (3) *The error that remains is the single site's, and it is the projection*
 75 *error of the medium.* The single-site T -matrix sums the rescattering within a
 76 voxel to all orders, and the coupling between voxels is exact, so the multiple
 77 scattering itself is not approximated. But the T -matrix is that of the voxel as
 78 represented: its contrast projected to degree r and its field held to degree p .
 79 When the scattered field is separated by its order in the contrast, the error
 80 of the second-order term, in which the field scattered once within a voxel
 81 is scattered there again, equals the *relative projection error* of the medium:
 82 the fraction of the squared norm of its profile that polynomials of degree
 83 $\min(p, r)$ on the cells cannot represent (Eq. (54)). The equality is exact in
 84 the long-wave limit, where it is derived, and holds to within 1.3% for all nine
 85 pairs with $p, r \leq 2$ in the experiments (§9). Splitting that term confirms
 86 whose error it is: the part in which both scatterings lie in one voxel carries
 87 the error, and the part between voxels at most 0.2% of it. It shows what
 88 the wavefield basis is for: it must carry what the medium imprints on the
 89 field, not only the wave, which at these frequencies varies little across a cell.
 90 (4) *The two bases must be matched, and the error is known in advance.* The
 91 relative projection error is a property of the medium and the grid alone, so

92 the error of a calculation is known before anything is solved, and the cell size
 93 and the degree that a required accuracy needs can be chosen beforehand.
 94 (5) *Where the medium is piecewise constant on the cells the projection error*
 95 *vanishes*, a constant suffices for the medium, and the degree of the field
 96 alone sets the order: in a randomly stratified model one first-moment voxel
 97 per model layer reaches a reflection accuracy of 10^{-6} , where the uniform-
 98 field voxel needs about forty (§9), and in the uniform layer it allows voxels
 99 58 times larger in each direction (§12). (6) *In three dimensions* the smoothly
 100 graded sphere converges at second order with the mean-only voxel, as the
 101 layer predicts (§10), and at fourth order with the first-moment voxel in the
 102 companion paper [4].

103 *What makes the experiment exact.* The answer above can be trusted only if
 104 nothing but the two bases is approximated. Three results, set against the
 105 coupled-dipole method described below, secure that. (a) *An exact coupling.*
 106 The integrated Green tensor and the filtered coupled dipoles approximate the
 107 field of a cell; for cubes the source-cell average is exact, because the cube's
 108 form factor vanishes on the reciprocal lattice (the tiling identity, §5), and
 109 at normal incidence the lattice kernel is in closed form with no lattice sum.
 110 (b) *No polarisability to choose.* The polarisability, a modelling choice in the
 111 coupled-dipole method, is here a derived object (§6): the moments of the
 112 Green's tensor over the cube, taken as distributions, yield Eshelby's δ term
 113 without adding it by hand, the cube's departure from the sphere as a pure
 114 cubic anisotropy, and the closure's error against exact scattering by a sphere
 115 as a real $(ka)^2$ term in closed form. The self term the single site carries is
 116 the one the averaged kernel omits, so the discrete system is a collocation of

117 the continuum equation. (c) *The elastic state.* A voxel carries a force and
118 a moment, so its state has nine components, the coupling is a 9×9 tensor,
119 and its local term is the depolarisation of a plate.

120 *Why cubic voxels.* Cubes are the natural cell for three reasons. They *tile*
121 *space*, so a uniform region discretised into cubes is still that uniform region,
122 whereas spheres or point scatterers can only approximate it and carry a pack-
123 ing error of their own. Their *regular lattice* makes the interaction between
124 cells translation invariant, so the coupling is a convolution (Toeplitz between
125 planes, Bloch-periodic within them), and the whole system is solved by fast
126 transforms. And a cube small compared with the wavelength has a *single-site*
127 *response in closed form*, the elastic depolarisation of a cube (§6), so the only
128 approximation a voxel introduces is the internal field it assumes.

129 *The coupled-dipole method.* Replacing a heterogeneous body by a lattice of
130 small scatterers that interact through the background Green’s function is the
131 multiple-scattering description of Foldy [5] and Lax [6]. In electromagnetics
132 it became a numerical method as the coupled-dipole method, or discrete
133 dipole approximation (DDA): each cell of a cubic grid is a polarisable point
134 dipole driven by the incident field and by the fields of every other dipole
135 [7, 8]. The method is a collocation of the volume integral equation for the
136 electric field, with the cell’s self-interaction absorbed into its polarisability
137 [9], and Yurkin and Hoekstra [10] review it in that form. Its accuracy turns
138 on two ingredients, and this paper takes up both for elasticity. The first is the
139 polarisability. The Clausius–Mossotti value is the static response of a sphere;
140 the lattice dispersion relation of Draine and Goodman [11] adjusts it so that
141 an infinite lattice propagates waves as the continuum does to a given order

142 in kd ; Rahmani et al. [12] instead derive it from the exact static solution for
 143 the whole body, so that the method is exact in the long-wavelength limit.
 144 The second is the coupling. Integrating the Green tensor over the source
 145 cell rather than sampling it at the centre improves the accuracy, most of
 146 all for strong contrasts [13], and the filtered coupled dipoles of Piller and
 147 Martin [14], revived by Yurkin et al. [15] for large refractive indices, replace
 148 the point dipole by a source of finite extent whose spectrum is band-limited
 149 to the grid. The coupling summed over a lattice extends the method to
 150 periodic structures [16]. Its error converges at second order in the dipole
 151 spacing when shape errors are absent [3], and for a cube, the one shape
 152 a cubic grid represents without a staircase, Yurkin and Kahnert [17] reach
 153 relative uncertainties of 10^{-7} to 10^{-3} by extrapolation in the grid. On the
 154 mathematical side, Costabel et al. [18] show that the method is a finite section
 155 of an infinite block Toeplitz matrix, stable for a large range of the contrast
 156 but not for all of it, and Costabel [19] that in a one-dimensional model it
 157 converges at second order in the interior, with a boundary layer in which
 158 the consistency error does not vanish. The layer studied here is the elastic
 159 counterpart of that cube: a body the grid represents exactly, so that the
 160 discretisation error can be isolated. What this paper adds is the closed form:
 161 for cells that tile space, the coupling that makes the lattice reproduce the
 162 continuum exactly, and the order and the constant of the error that remains,
 163 as functions of the two bases. In the terms used here the coupled-dipole
 164 method is the pair $p = r = 0$, and its refinements improve the polarisability
 165 and the coupling while leaving both bases constant in the cell.

166 *Its elastic counterparts.* The elastic problem has the same volume integral
 167 equation [20], and every cell-based solver of it is a coupled-dipole method
 168 whether or not it carries the name. Nestor [21] divided a two-and-a-half-
 169 dimensional medium into cells that interact through the background Green's
 170 tensor, each carrying a contrast renormalised by its own near field, with
 171 the multiple-scattering series summed. Kanaun and Levin [22] discretise the
 172 inclusion problem with Gaussian approximating functions whose matrix ele-
 173 ments are analytic. Yang et al. [23] solve a stress-velocity form of the integral
 174 equation on a grid by conjugate gradients with fast Fourier transforms, and
 175 Touhei [24] does the same for a half space with a generalised Fourier trans-
 176 form. Jakobsen [25] and Jakobsen et al. [26] build the seismic forward and
 177 inverse problems on the transition operator of the discretised Lippmann-
 178 Schwinger equation, in the anisotropic elastic case with a nine-component
 179 state vector. The elastic problem adds a structure the scalar and electro-
 180 magnetic ones lack. A cubic voxel responds to the incident field with a force
 181 (through its density contrast) and a moment (through its stiffness contrast),
 182 so the state at a voxel has nine components (displacement and strain), and
 183 the single site's self-interaction is the elastic depolarisation of a cube, the
 184 discrete counterpart of the inclusion tensor of Eshelby [27].

185 *This paper.* It extends the formulation of Nestor [21] to cubic voxels, and is
 186 ordered as the experiment is built. §2 sets out the discrete medium and §3 the
 187 continuous layer it must reproduce. §4 shows why the point coupling fails,
 188 and §5 why the source-averaged coupling succeeds exactly. §6 derives the
 189 single-site T -matrix from moment integrals, as the first member of a hierarchy
 190 in the gradients of the field whose moments are all in closed form, shows what

191 each further block of that hierarchy buys, tests it against the exact sphere
 192 and on a cube, and shows that with its first member the discrete system is
 193 a collocation of the continuum equation. The wavefield basis comes next: §7
 194 gives the second-order error of the uniform field in closed form, and §8 the
 195 voxel whose field carries Legendre moments to degree p , with its order $2p+2$
 196 and its constant, for incident P and S at normal and oblique incidence. The
 197 medium basis follows in §9: a randomly stratified layer, for which a constant
 198 in each cell is exact, and a medium that varies within a cell, for which the
 199 contrast is expanded to degree r , the order becomes $\min(2p+2, 2r+2)$ and
 200 the error of the single-site T -matrix is the relative projection error of the
 201 medium. §10 carries the prediction to a sphere, in its far field and its near
 202 field, and §11 builds the hierarchy of §6 into a voxel scheme and sets it beside
 203 the Legendre voxels. §12 discusses what the results imply for voxel schemes in
 204 general, and §13 closes with why an error known in advance is worth having.
 205 Every result is derived symbolically in Mathematica and checked against an
 206 independent Python implementation or an exact solution.

207 2. Setting

208 *The discrete medium.* Space is tiled by cubes of side d (half-side $h = d/2$).
 209 Cube i carries a contrast $\Delta\lambda, \Delta\mu, \Delta\rho$ from an isotropic background with
 210 Lamé parameters λ, μ , density ρ , P and S speeds α, β and P-wave modulus
 211 $M_P = \lambda + 2\mu$. Time dependence is $e^{-i\omega t}$; $k_P = \omega/\alpha$, $k_S = \omega/\beta$. Coordinates
 212 are ordered (z, x, y) with z down. The field at a voxel is the nine-component
 213 state

$$\psi = (u_z, u_x, u_y, \varepsilon_{zz}, \varepsilon_{xx}, \varepsilon_{yy}, 2\varepsilon_{xy}, 2\varepsilon_{zy}, 2\varepsilon_{zx}), \quad (1)$$

214 displacement followed by engineering strain. A voxel responds with a force
 215 and a moment (stress dipole); the 9×9 single-site T -matrix maps the exciting
 216 state to them, and the 9×9 lattice Green's tensor $\mathbf{K}(i, j)$ maps a source
 217 voxel's force and moment to the state at voxel i . The Foldy–Lax system is

$$\psi_i = \psi_i^0 + \sum_{j \neq i} \mathbf{K}(i, j) \mathbf{T}_j \psi_j, \quad (2)$$

218 with ψ^0 the incident state.

219 *Where the kernel comes from.* Write the medium as the background plus
 220 the contrast. With time dependence $e^{-i\omega t}$ the equation of motion is $\partial_c \sigma_{bc} +$
 221 $\rho \omega^2 u_b = 0$, with $\sigma_{bc} = c_{bcde} \varepsilon_{de}$ and summation over repeated indices. Split-
 222 ting $\rho = \rho_0 + \Delta\rho$ and $c = c^0 + \delta c$, with $\delta c_{bcde} = \Delta\lambda \delta_{bc} \delta_{de} + \Delta\mu (\delta_{bd} \delta_{ce} + \delta_{be} \delta_{cd})$
 223 the contrast stiffness, and moving the contrast terms to the right leaves the
 224 background operator on the left and the contrast on the right as a body force,

$$f_b = \omega^2 \Delta\rho u_b + \partial_c m_{bc}, \quad m_{bc} = \delta c_{bcde} \varepsilon_{de}, \quad (3)$$

225 a force from the density contrast, and the divergence of the contrast stress m ,
 226 the moment, from the stiffness contrast. With $\mathbf{G}$ the Green's tensor of the
 227 background, the displacement is $u = u^0 + \int \mathbf{G} f \, d\mathbf{y}$, and the moment term
 228 integrates by parts,

$$\int G_{ab}(\mathbf{x}-\mathbf{y}) \frac{\partial m_{bc}}{\partial y_c} \, d\mathbf{y} = - \int \frac{\partial G_{ab}}{\partial y_c}(\mathbf{x}-\mathbf{y}) m_{bc} \, d\mathbf{y} = + \int \frac{\partial G_{ab}}{\partial x_c}(\mathbf{x}-\mathbf{y}) m_{bc} \, d\mathbf{y}, \quad (4)$$

229 the boundary term vanishing because m is confined to the scatterers, and the
 230 last step holding because $\mathbf{G}$ depends only on the separation. The moment
 231 therefore radiates through the *receiver* derivative of $\mathbf{G}$, with a plus sign, when

232 the moment is the contrast stress itself. With the moment tensor defined
 233 instead through its equivalent body force $-\partial_c[M_{bc}\delta(\mathbf{x}-\mathbf{y})]$, as in Aki and
 234 Richards [28], the same field reads $M_{bc}\partial G_{ab}/\partial y_c$, the source derivative, with
 235 $M = -m$. The two are one convention written two ways; what matters is that
 236 the derivative in the kernel and the sign in the contrast operator are chosen
 237 together. The nine-component kernel follows (Fig. 1): its force columns are
 238 $\mathbf{G}$ and its moment columns the receiver derivatives $\partial_c G_{ab}$, symmetrised over
 239 (b, c) with a factor $\frac{1}{2}$ on the shear components; its displacement rows are
 240 those entries and its strain rows their receiver derivatives, symmetrised in
 241 the same way and doubled on the shear rows to give engineering strain. In
 242 that convention the contrast operator is

$$\Delta = \text{blockdiag}(\omega^2 \Delta \rho \mathbf{I}_3, \Delta \mathcal{C}), \quad (5)$$

243 where $\Delta \mathcal{C}$ is the 6×6 Voigt form of δc , the contrast stiffness with $\Delta \lambda + 2\Delta \mu$
 244 and $\Delta \lambda$ in the normal block and $2\Delta \mu$ (not $\Delta \mu$) on the shear diagonal.

245 *Averaging, and collocation.* Three operations in this paper are averages over
 246 a cell, and they are distinct. The *source-cell average* integrates the kernel
 247 over the cell that radiates: $\langle \mathbf{K} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{K}(\mathbf{r} - \mathbf{y}) d\mathbf{y}$ is the field at $\mathbf{r}$
 248 of a uniform polarisation of the source cell. Every scheme in this paper uses
 249 it, and §5 shows it to be exact. *Collocation* fixes where the discrete equation
 250 is imposed: at the receiving cell's centre, with the internal field of each cell
 251 uniform, so that the unknown is the field at the centre. The *receiver-cell*
 252 *average* imposes the equation in the mean over the receiving cell instead;
 253 with a uniform internal field that is the Galerkin scheme of degree zero,
 254 called the mean-only voxel below. The two are both *uniform-field* voxels, a

term that refers to the internal field and not to the test, and they differ only in the test function (a delta at the centre against the cell’s indicator) and in the constant of their $O((kd)^2)$ error (Table 7). The *cell-mean contrast*, finally, is an average of the material, not of the field: where the medium varies within a cell, the contrast is projected onto the cell’s basis, its mean for degree $r = 0$, rather than sampled at the centre (§§9, 10). In the language of the coupled-dipole method, the point dipole is collocation with the point kernel, the integrated Green tensor adds the source-cell average, and the filtered coupled dipoles replace the point source by one of finite extent.

The laterally uniform layer. A layer of thickness D is filled with n planes of voxels, $d = D/n$, and illuminated at normal incidence. The field is then laterally uniform, only the zero-lateral-wavenumber (specular) component of the lattice coupling acts, and Eq. (2) becomes exactly a chain of n planes coupled by the specular lattice sums

$$\mathbf{S}(m) = \sum_{\mathbf{R}} \mathbf{G}(\mathbf{R} + md\hat{\mathbf{z}}), \quad (6)$$

where the sum runs over the lattice vectors $\mathbf{R} = d(0, p, q)$, p, q integers, that join the centres of the voxels in one plane (a square lattice of pitch d), and m is the separation of the two planes in units of d (Fig. 2a,b). The question is whether, and how fast, this chain converges to the continuous layer as $n \rightarrow \infty$ at fixed D .

Implementations. Every result is derived symbolically in Mathematica and compared with an independent numerical implementation or an exact solution. “The package” below is the Python implementation of the voxel scheme

277 (`cubic_scattering`): its cube T -matrix, its lattice kernels and its stratified
 278 propagator, which serve both as the object under test and, where validated,
 279 as a second implementation.

280 3. The continuous layer and its thin-layer expansion

281 At normal incidence a plane force drives a one-dimensional problem per
 282 wave type, with modulus M (M_P for P, μ for S), wavenumber k and plane
 283 Green's function $g(z) = \frac{i}{2Mk} e^{ik|z|}$. The layer's reflection and transmission
 284 follow from continuity of displacement and traction at its faces; they agree
 285 with the stratified propagator used as the reference in this work to 2.5×10^{-8} ,
 286 P and S (`ContinuumLimit_Reference.wl`).

287 For a layer of thickness D with modulus M_1 and density ρ_1 in a back-
 288 ground M_0, ρ_0 , write $Z_j = \sqrt{M_j \rho_j}$ for the impedances, $\zeta = Z_1/Z_0$ for their
 289 ratio and $k_1 = \omega \sqrt{\rho_1/M_1}$ for the layer's wavenumber. The reflection coeffi-
 290 cient expands as

$$R = \frac{ik_1 D}{2} (\zeta - \zeta^{-1}) + \frac{(k_1 D)^2}{4} (\zeta^{-2} - \zeta^2) - \frac{i(k_1 D)^3}{24} (\zeta - \zeta^{-1}) (3\zeta^{-2} + 2 + 3\zeta^2) \\ + \frac{(k_1 D)^4}{48} (\zeta^2 - \zeta^{-2}) (3\zeta^2 - 2 + 3\zeta^{-2}) + O(D^5), \quad (7)$$

291 exact in the contrast (`ContinuumLimit_ThinLayer.wl`, which checks each
 292 coefficient symbolically and the truncated series against the exact layer).
 293 The first term is

$$\frac{i\omega D}{2} \sqrt{\frac{M_0}{\rho_0}} \left[\frac{\rho_1 - \rho_0}{M_0} - \rho_0 \left(\frac{1}{M_1} - \frac{1}{M_0} \right) \right] : \quad (8)$$

294 to first order a thin layer is an interface across which the traction jumps by
 295 the excess inertia and the displacement by the excess compliance. At oblique

296 incidence the same statement holds with the first-order system $\partial_z \mathbf{b} = \mathbf{A} \mathbf{b}$
 297 for the P-SV state $\mathbf{b} = (u_x, u_z, t_{xz}, t_{zz})$: the layer acts to $O(D)$ as the jump
 298 $\mathbf{I} + (\mathbf{A}_1 - \mathbf{A}_0)D$, which reproduces the layer's $O(D)$ reflection coefficients
 299 R_{PP}, R_{PS} to 10^{-12} ; the series truncated at $O(D^k)$ converges at exactly order
 300 k ; and a soft thin layer with shear modulus $\mu_1 = D/\eta$ at fixed compliance η
 301 tends to the linear-slip interface as $D \rightarrow 0$. These expansions are the target
 302 a single plane of voxels must reproduce order by order. A plane of voxels
 303 that carry Legendre moments of their internal field to degree p reproduces
 304 Eq. (7) term by term through $O(D^{2p+2})$ and first differs at D^{2p+3} (§8): the
 305 mean-only voxel reproduces the equivalent interface and the D^2 term, the
 306 first-moment voxel all four terms shown.

307 4. The specular sums of the point Green's tensor

308 *Why transform the sum.* The specular sum (6) adds, at a receiver, the field
 309 of every voxel of a plane m planes away. In that form it is a poor object to
 310 study. The Green's tensor decays slowly with distance, so the sum converges
 311 slowly, and it hides the one question that matters here: which part of it is
 312 the continuum, and which part is the discretisation. The Poisson summation
 313 formula [29] answers both at once, and is the standard tool for lattice sums
 314 of wave fields [30]. For a function f on the plane and the square lattice of
 315 pitch d it states

$$\sum_{\mathbf{R}} f(\mathbf{R}) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{f}(\mathbf{g}), \quad \hat{f}(\mathbf{g}) = \int f(\mathbf{x}) e^{-i\mathbf{g} \cdot \mathbf{x}} d^2x, \quad \mathbf{g} = \frac{2\pi}{d}(p, q),$$

(9)

316 with p, q integers: a sum of samples of f on the lattice equals a sum of samples
 317 of its Fourier transform on the reciprocal lattice, whose pitch is $2\pi/d$ (Fig. 3).

318 Applied to the specular sum,

$$\mathbf{S}(m) = \frac{1}{d^2} \sum_{\mathbf{g}} \hat{\mathbf{G}}(\mathbf{g}; md), \quad (10)$$

319 with $\hat{\mathbf{G}}(\mathbf{g}; z)$ the two-dimensional (lateral) transform of the Green's tensor
 320 between planes a distance z apart. Each term has a physical meaning. Trans-
 321 formed laterally, the field of a plane of sources is a superposition of plane
 322 waves, one for each lateral wavenumber, and the lattice selects the wavenum-
 323 bers $\mathbf{g}$. The order $\mathbf{g} = 0$ is the laterally uniform plane wave that a continuous
 324 layer radiates: it is the continuum's term. Every other order has $|\mathbf{g}| \geq 2\pi/d$,
 325 far beyond the wavenumber of the field when the voxels are small compared
 326 with the wavelength, so it is evanescent and decays away from the plane as
 327 $e^{-|\mathbf{g}|md} = e^{-2\pi m \sqrt{p^2+q^2}}$, independently of d . The transform therefore splits
 328 the lattice sum exactly into the continuum and a discretisation correction,
 329 and the correction converges after a few orders.

330 *The local term that survives.* In the strain-from-moment block the evanescent
 331 orders carry two powers of $|\mathbf{g}| \propto 1/d$, so their contribution per voxel volume
 332 does *not* vanish as $d \rightarrow 0$: at $m = 1$ it is eleven times the continuum
 333 term (`ContinuumLimit_SpecularSums.wl`, matched to the package's Ewald
 334 lattice sum to 5×10^{-13}). This surviving term is static. Summed over all
 335 planes in slab order, the static lattice sum of the point kernel is exactly
 336 the depolarisation of a uniform plate ($-1/M_P$ on $\varepsilon_{zz} \leftarrow M_{zz}$ and $-1/(2\mu)$
 337 on the normal shears, per unit volume) plus a cubic-symmetric remainder
 338 (`ContinuumLimit_StaticLocal.wl`, to 2×10^{-10}). The point kernel therefore
 339 reproduces the continuum only if that remainder is cancelled by the voxel's
 340 own self term.

341 5. The tiling identity

342 Replace the point coupling by the *source-cell average* $\langle \mathbf{G} \rangle(\mathbf{r}) = d^{-3} \int_{\text{cube}} \mathbf{G}(\mathbf{r} -$
 343 $\mathbf{y}) d\mathbf{y}$, the field at a voxel centre of a uniform polarisation of the source voxel.
 344 Each Poisson order in Eq. (10) is then multiplied by the cube's form factor

$$F(\mathbf{g}) = \text{sinc}(g_x h) \text{sinc}(g_y h), \quad (11)$$

345 which vanishes at every $\mathbf{g} \neq 0$ of the reciprocal lattice since $\text{sinc}(p\pi) = 0$ for
 346 integer $p \neq 0$. This is the Fourier form of the fact that cubes tile the plane,
 347 $\sum_{\mathbf{R}} \chi(\mathbf{x} - \mathbf{R}) = 1$ for the cube's indicator χ : a partition of unity (Fig. 4).
 348 Two exact statements follow.

349 *Between planes.* For $m \neq 0$ every evanescent order is removed and

$$\mathbf{S}_{\text{avg}}(m) = \frac{1}{d^2} \sum_{W=P,S} \hat{\mathbf{G}}_W(0; md) \text{sinc}(k_W h), \quad (12)$$

350 the continuum's plane-wave term times the vertical form factor. The pack-
 351 age's cell-averaged kernel agrees in all 81 entries to 2.8×10^{-16} (`ContinuumLimit_`
 352 `AveragedSums.wl`).

353 *Within a plane.* The sum over *all* cells of the plane, the voxel's own included,
 354 is the field at the mid-plane of a uniform plate of thickness d (Fig. 2c):

$$d^3 \mathbf{S}_{\text{avg}}(0) + \mathbf{S}_{\text{self}} = d^3 \mathbf{P}_0, \quad \mathbf{S}_{\text{self}} = \int_{\text{cube}} \mathbf{G}(-\mathbf{y}) d\mathbf{y}, \quad (13)$$

355 where $\mathbf{P}_0 = d^{-3} \int_{-h}^h \mathbf{G}_{\text{plate}}(-z') dz'$ is elementary: in the one-dimensional
 356 transform each entry is $c_0 + A/(Mk_z^2 - \rho\omega^2)$, so it averages to $c_0/d + A(e^{ikh} -$
 357 $1)/(dMk^2)$. Its only nonzero entries are $u \leftarrow f$ on the diagonal and the
 358 three normal strain-from-moment entries, whose local weights $-1/M_P$ and

359 $-1/(2\mu)$ are the plate depolarisation of §4. Statically the identity holds for
 360 all 21 independent components of the strain-from-moment tensor to 4×10^{-9} ,
 361 with every cell integral evaluated analytically (below).

362 *Consequence: an exact kernel with no lattice sum.* At normal incidence the
 363 same-plane block of the cell-averaged Foldy–Lax kernel is

$$\mathbf{S}_{\text{avg}}(0) = \mathbf{P}_0 - \mathbf{S}_{\text{self}}/d^3, \quad (14)$$

364 with no Ewald splitting, near shell or asymptotic tail. The route it replaces
 365 (Ewald summation with a near shell averaged by quadrature and an $O(d^2)$
 366 Taylor tail beyond it) carries a static strain-from-moment error of 9.4×10^{-4}
 367 at its default settings, falling to 2.6×10^{-5} only when both the shell radius
 368 and the quadrature order are raised; Eq. (14) is now the package’s kernel at
 369 normal incidence.

370 *Cell integrals without quadrature.* Write the static (Kelvin) tensor as $G_{bc} =$
 371 $(c_A + c_B) \delta_{bc}/r - c_B \partial_b \partial_c r$ with $c_B = (1/\mu - 1/M_P)/8\pi$ and $c_A = 1/(4\pi\mu) - c_B$.
 372 Every cell integral of its derivatives is a cell integral of derivatives of $1/r$ and
 373 r . One application of the divergence theorem moves it onto faces $x_p = c_p \pm h$,
 374 which never contain the origin, and there the two-dimensional antiderivative
 375 in the face coordinates is continuous, so the face integral is a sum over the
 376 face’s corners. The distributional term at $r = 0$ is carried by the divergence
 377 theorem itself. (A triple antiderivative evaluated at the cube’s corners is
 378 *not* usable for repeated derivatives: its arctangent terms jump across the
 379 coordinate planes through the origin, and differentiating twice there discards
 380 exactly that term.) The self term so obtained reproduces the classical closed
 381 form of the cube’s depolarisation to machine precision.

382 6. The single site: the cube T -matrix from moment integrals

383 Every error of a voxel scheme is made at the single site (§12), so the single
384 site is derived here rather than imported. The classical route imports it. For
385 an ellipsoid, Eshelby’s uniformity theorem makes the internal strain under a
386 uniform load uniform, with a depolarisation tensor in closed form [27, 31, 32];
387 its dynamic generalisation is known for spheres and ellipsoids [33, 34]. For
388 a cube or a general polyhedron the static field of a uniform eigenstrain is
389 also in closed form [35, 36, 37, 38], but the strain it produces is not uni-
390 form, so there is no exact uniform-strain inclusion to import. Closed-form
391 Eshelby tensors for polyhedral inclusions with polynomial eigenstrains have
392 since been given [39], the dynamic inclusion problem has been formulated
393 for shapes other than the ellipsoid, cubes among them [40], and the Green’s
394 tensor has been integrated analytically over a cuboid for the coupled-dipole
395 method [41]; these are the static and electromagnetic counterparts of in-
396 dividual moments below. In scattering theory the same object appears as
397 the self-consistent or effective-field closure of the volume integral equation
398 [42, 20, 43]: a first-order (Born) scatterer [44] renormalised by the geometric
399 series of its own near field, the step that turns a contrast into an effective
400 contrast. Nestor [21] introduced the single site in this form, summing the
401 series of the cell’s self integral, with the singular near field evaluated over an
402 infinitesimal cell. The moments below evaluate the same self integral over
403 the finite cube, as a distribution. The elastic T -matrix of Waterman [45]
404 represents a scatterer in spherical multipoles, the natural basis for a sphere;
405 a voxel in a lattice needs one compatible with the lattice, and the moments
406 over the cube provide it. The construction below makes the approximation of

407 a voxel explicit rather than implicit: it derives the cube's T -matrix from the
 408 moments of the elastodynamic Green's tensor over the cube, and identifies
 409 the uniform-strain closure as one truncation of an exact hierarchy.

410 6.1. The closed hierarchy

411 Let the voxel be $V = [-h, h]^3$ with contrast $\delta c_{ijkl} = \Delta\lambda \delta_{ij}\delta_{kl} + \Delta\mu(\delta_{ik}\delta_{jl} +$
 412 $\delta_{il}\delta_{jk})$ and $\Delta\rho$. Expand the internal displacement about the centre to second
 413 degree and project the Lippmann–Schwinger equation onto 1, x_p and x_px_q .
 414 The unknowns are $u_i(0)$, $\partial_p u_i(0)$ and $\partial_p \partial_q u_i(0)$, and the result is the closed
 415 system

$$(\delta_{ij} - \omega^2 \Delta\rho \mathbf{G}_{ij})u_j + (\mathbf{N}_{in,k}^r \delta c_{nksj} - \frac{1}{2}\omega^2 \Delta\rho \mathbf{K}_{ij}^{rs})\partial_r \partial_s u_j = u_i^0, \quad (15)$$

$$(\delta_{pr}\delta_{ij} + \omega^2 \Delta\rho \mathbf{N}_{ij,p}^r - \mathbf{M}_{in,pk} \delta c_{nkrj})\partial_r u_j = \partial_p u_i^0, \quad (16)$$

$$-\omega^2 \Delta\rho \mathbf{M}_{ij,pq} u_j + (\delta_{qr}\delta_{ps}\delta_{ij} + \mathbf{Q}_{in,pqk}^r \delta c_{nksj} - \frac{1}{2}\omega^2 \Delta\rho \mathbf{P}_{ij,pq}^{rs})\partial_r \partial_s u_j = \partial_p \partial_q u_i^0, \quad (17)$$

416 all fields evaluated at the centre. Its coefficients are six moments of the
 417 background Green's tensor $\mathbf{G}$ over the voxel,

$$\begin{aligned} \mathbf{G}_{ij} &= \int_V G_{ij} \, dV, & \mathbf{K}_{ij}^{rs} &= \int_V G_{ij} x_r x_s \, dV, & \mathbf{N}_{in,k}^r &= \int_V \partial'_k G_{in} x_r \, dV, \\ \mathbf{M}_{in,pk} &= \int_V \partial'_p \partial'_k G_{in} \, dV, & \mathbf{P}_{ij,pq}^{rs} &= \int_V \partial'_q \partial'_p G_{ij} x_r x_s \, dV, & \mathbf{Q}_{in,pqk}^r &= \int_V \partial'_q \partial'_p \partial'_k G_{in} x_r \, dV, \end{aligned} \quad (18)$$

418 with derivatives taken with respect to the source point, $\partial' = -\partial$. That sign
 419 reaches $\mathbf{N}$ and $\mathbf{Q}$ only, and no symmetry, parity or scaling check can detect
 420 it, so it is fixed by convention. The moments are graded by (D, W) , the

421 number of derivatives and the degree of the weight: $\mathbf{G}(0, 0)$, $\mathbf{K}(0, 2)$, $\mathbf{N}(1, 1)$,
 422 $\mathbf{M}(2, 0)$, $\mathbf{P}(2, 2)$, $\mathbf{Q}(3, 1)$.

423 *The hierarchy at any degree.* Nothing in the construction is special to the
 424 second degree. Let the internal displacement be expanded to degree ℓ ,

$$u_j(\mathbf{x}) = \sum_{|W| \leq \ell} \frac{x_W}{|W|!} U_{j;W}, \quad U_{j;W} = \partial_W u_j(0), \quad x_W = x_{w_1} \cdots x_{w_m}, \quad (19)$$

425 the sum running over ordered sets of indices $W = (w_1, \dots, w_m)$, and let the
 426 strain be its derivative. Write the moments with any number of derivatives
 427 and any weight as

$$\mathbf{g}_{in}[D; W] = \langle \partial_D G_{in}, x_W \mathbf{1}_V \rangle, \quad (20)$$

428 the derivatives being those of G_{in} as a function of its argument and the
 429 pairing distributional (§6.2); the six moments of Eq. (18) are those with
 430 $|D| + |W| \leq 4$, up to the sign $(-1)^{|D|}$ of the source-derivative convention.
 431 Imposing the Lippmann–Schwinger equation and its derivatives to order ℓ at
 432 the centre gives, for every set of derivative indices P with $|P| \leq \ell$,

$$U_{i;P} - (-1)^{|P|} \omega^2 \Delta \rho \sum_{|W| \leq \ell} \frac{\mathbf{g}_{ij}[P; W]}{|W|!} U_{j;W} + (-1)^{|P|} \sum_{|W| \leq \ell-1} \frac{\mathbf{g}_{in}[Pk; W]}{|W|!} \delta c_{nkrj} U_{j;rW} = \partial_P u_i^0(0). \quad (21)$$

433 Equations (15)–(17) are the case $\ell = 2$, and the uniform-strain closure is
 434 $\ell = 1$. The cube is centrosymmetric, so a moment vanishes unless $|D| + |W|$
 435 is even, and the system needs the grades

$$\begin{aligned} (D, W) &= (m, W), \quad W \leq \ell && \text{(density),} \\ (D, W) &= (m+1, W), \quad W \leq \ell-1 && \text{(stiffness),} \end{aligned} \quad m = 0, \dots, \ell, \quad D+W \text{ even.} \quad (22)$$

436 Table 1 lists what each degree adds. The unknowns number $3\binom{\ell+3}{3}$: 12, 30,
 60 for $\ell = 1, 2, 3$.

Table 1: The grades (D, W) of moments, D derivatives on the Green's tensor and a weight
 of degree W , that the hierarchy of degree ℓ needs beyond those of degree $\ell - 1$.

ℓ	new grades	moments in all
1	(0, 0), (1, 1), (2, 0)	3
2	(0, 2), (2, 2), (3, 1)	6: the set of Eq. (18)
3	(1, 3), (3, 3), (4, 0), (4, 2)	10
4	(0, 4), (2, 4), (4, 4), (5, 1), (5, 3)	15
5	(1, 5), (3, 5), (5, 5), (6, 0), (6, 2), (6, 4)	21

437

438 6.2. The moments as distributions

439 Write the elastodynamic tensor as

$$G_{in}(\mathbf{r}) = \frac{1}{4\pi\mu} \left[\delta_{in} g_S(r) + \frac{1}{k_S^2} \partial_i \partial_n (g_S(r) - g_P(r)) \right], \quad g_c(r) = \frac{e^{ik_c r}}{r} \quad (23)$$

440 [46, 28]. Expanding g_c in powers of r reduces every moment in Eq. (18) to a
 441 finite sum of scalar moments

$$E[m; d_1 \dots d_D; w_1 \dots w_W] = \langle \partial_{d_1} \dots \partial_{d_D} r^m, x_{w_1} \dots x_{w_W} \mathbf{1}_V \rangle. \quad (24)$$

442 For $m = -1$ and $D \geq 2$ the classical integrand is not absolutely integrable.
 443 $\partial^2(1/r)$ decays as r^{-3} , its angular average vanishes, and the value of the inte-
 444 gral depends on the shape of the region excluded around the origin. This is
 445 the source of the δ -function term of the inclusion problem, which a numerical
 446 quadrature misses. Without it the depolarisation is wrong; for a sphere it
 447 vanishes altogether (§6.3). It is not added afterwards here. The derivatives

are distributional from the start, and each is moved onto the voxel's faces by pairing with the test function $w\mathbf{1}_V$, using $\partial_q\mathbf{1}_V = -n_q\delta_{\partial V}$:

$$\langle \partial_q T, w\mathbf{1}_V \rangle = \oint_{\partial V} n_q w T \, dA - \langle T, (\partial_q w) \mathbf{1}_V \rangle. \quad (25)$$

The surface term is evaluated on the faces, a distance h from the origin, where every derivative of r^m is smooth. The recursion lowers D and W together and ends on convergent integrals. The face integrals are done in closed form by the corner sums of §5. The sharp check is the sum rule

$$\sum_p E[-1; p, p;] = \nabla^2(1/r) \text{ paired with } \mathbf{1}_V = -4\pi, \quad (26)$$

which the face route satisfies exactly and a spherical excision returns as 0. The closed forms contain logarithms of the units $(2 \pm \sqrt{3})^n$ of the Pell equation $x^2 - 3y^2 = 1$. Left unreduced, these cancel catastrophically: for example, $\log(72010600134783751 - 41575339372323900\sqrt{3})$ is a cancellation of 35 digits. They are therefore reduced to $n \log(2 - \sqrt{3})$, which is exact. Because every power of r is paired this way, the dynamic moments keep the even-power terms of the expansion $(k^2r, r^3, \dots)$ as well as the odd ones. A route that integrates only the polynomial (odd-power) terms of $\mathbf{G}$ misses those terms, at a real $O((kh)^2)$ cost (§6.6).

Every grade, in closed form on four constants. The recursion (25) does not depend on the grade, so the same engine gives every scalar moment of Table 1. Each is evaluated once and stored. The scalar moments of every grade through $\ell = 5$, with up to six derivatives and weights to the fifth degree, have been evaluated for the kernels r^{-1} and r : 4224 closed forms. (The second term of Eq. (23) places two further derivatives on r ; those moments

are stored through $\ell = 3$ and are evaluated numerically, by the second route below, beyond it.) Every one of them is a power of the voxel's width times

$$q_0 + q_1 \sqrt{3} + q_2 \pi + q_3 \log(2 - \sqrt{3}), \quad q_i \text{ rational.} \quad (27)$$

The integration returns many of them in other forms, with $\log(\sqrt{3} - 1)$, $\log(1 + \sqrt{3})$, inverse hyperbolic tangents and logarithms of powers of two, which are equal in value but not canonical: two equal moments can then differ in appearance, and sums of large logarithms that cancel lose digits. Each stored form is therefore reduced to Eq. (27) by an integer-relation search at 400 digits, with $\log 2$ and $\log 3$ admitted to the search, and the reduction is accepted only if it agrees with the stored form to 1200 digits. All 4224 reduce, and the coefficients of $\log 2$ and $\log 3$ are zero in every one.

Two checks, whose answers do not depend on the engine, gate every grade. Because $\nabla^2 r^m = m(m+1)r^{m-2}$ and $\nabla^2(1/r) = -4\pi\delta$,

$$\sum_p E[m; ppR; W] = m(m+1) E[m-2; R; W], \quad \sum_p E[-1; ppR; W] = -4\pi (-1)^{|R|} (\partial_R x_W)(0), \quad (28)$$

for any further derivatives R and any weight W . The second is the sum rule (26) at every grade: it tests the δ function of the inclusion problem and its derivatives, which is where a quadrature fails. Both hold for every grade of Table 1 with two or more derivatives, 910 traces of each kind; the grades with fewer, for which the integrand is integrable, agree with numerical integration over the cube.

The closed forms are also confirmed by a route that shares nothing with the faces of the cube. The cube is its inscribed ball plus the remainder. On the ball the derivatives are moved onto the sphere, where the surface

490 integrals of monomials are Gamma functions, and all the distributional con-
 491 tent is there; on the remainder the integrand is an ordinary function, in-
 492 tegrated radially in closed form and over directions through the six faces
 493 by Gauss–Legendre quadrature. This evaluation reproduces all 4224 stored
 494 values, the largest difference being 1.1×10^{-12} , and satisfies Eq. (28) by
 495 itself to 5×10^{-10} or better up to eight derivatives (`CubeMomentCore.wl`,
 496 `CubeMomentHigherGrades.wl`, `CubeMomentStore.wl`, `CubeMomentCanonical`.
 497 `wl`; `scripts/crosscheck_cube_moments_ball_shell.py`).

498 6.3. Parity, and the nine-component block

499 The cube is centrosymmetric. A moment that couples a field quantity of
 500 degree a to one of degree b therefore vanishes unless $a + b$ is even. u , ∂u and
 501 $\partial\partial u$ have degrees 0, 1 and 2, so Eq. (16) contains the first gradient alone: ∂u is
 502 determined by its own block and couples to neither u nor $\partial\partial u$. The leading
 503 scattering by a stiffness contrast is the stress dipole $\int_V \delta c : \nabla u \, dV$, which
 504 needs only ∂u . The leading far field of the stiffness contrast is therefore fixed
 505 by Eq. (16), and *no treatment of the second-gradient block (17) can change*
 506 *it*. What Eq. (17) contributes reaches the far field only through the $O((kh)^2)$
 507 coupling in Eq. (15). That is the order, and the origin (the variation of the
 508 field across the voxel), of the collocation error of §7, which the first-moment
 509 voxel of §8 removes, as does the hierarchy itself once its third-gradient block
 510 is kept (§6.4).

511 With $\Delta\rho = 0$, Eq. (16) is the 9×9 operator $\delta_{pr}\delta_{ij} - \mathbf{M}_{in,pk}\delta c_{nkrj}$, built
 512 from $\mathbf{M}$ alone. Under O_h the nine gradient components decompose as $A_{1g} \oplus$
 513 $E_g \oplus T_{1g} \oplus T_{2g}$, and the operator is diagonal in that basis. Its eigenvalues are

514 the inverse strain concentrations:

$$A_{1g} \text{ (dilatation)} : 1 + \frac{\Delta\lambda + \frac{2}{3}\Delta\mu}{\lambda + 2\mu}, \quad (29)$$

$$T_{2g} \text{ (off-diagonal shear)} : 1 + 2\Delta\mu S_{\text{off}}, \quad S_{\text{off}} = \frac{\pi(\lambda + 2\mu) - \sqrt{3}(\lambda + \mu)}{3\pi\mu(\lambda + 2\mu)}, \quad (30)$$

$$E_g \text{ (diagonal shear)} : 1 + 2\Delta\mu S_{\text{diag}}, \quad S_{\text{diag}} = \frac{3\sqrt{3}(\lambda + \mu) + 2\pi\mu}{6\pi\mu(\lambda + 2\mu)}. \quad (31)$$

515 The density channel is the displacement block of Eq. (15) with the second
 516 gradients dropped, an amplification $(1 - \omega^2\Delta\rho\mathbf{G})^{-1}$ with $\mathbf{G}$ the cube integral
 517 of $\mathbf{G}$.

518 *The cube against the sphere.* For a sphere, Eshelby's tensor gives a dilata-
 519 tional channel $1 + \Delta K/(\lambda + 2\mu)$, with $\Delta K = \Delta\lambda + \frac{2}{3}\Delta\mu$, and a single deviatoric
 520 channel $1 + 2\Delta\mu S_{\text{sph}}$, with $2\mu S_{\text{sph}} = 2(4 - 5\nu)/[15(1 - \nu)]$. The cube's di-
 521 latational channel, Eq. (29), is the sphere's exactly: its eigenvalue involves
 522 $\mathbf{M}$ only through the contraction $\int_V \partial_i \partial_n G_{in} dV$, and the static Navier equa-
 523 tion makes $\partial_i \partial_n G_{in} = -\delta(\mathbf{r})/(\lambda + 2\mu)$, whose integral does not depend on
 524 the shape. The cube's shear channels differ from each other. Their isotropic
 525 average, however, is the sphere's, also exactly:

$$\frac{1}{5}(3S_{\text{off}} + 2S_{\text{diag}}) = S_{\text{sph}}, \quad S_{\text{diag}} - S_{\text{off}} = \frac{(\lambda + \mu)(5\sqrt{3} - 2\pi)}{6\pi\mu(\lambda + 2\mu)} > 0. \quad (32)$$

526 The $\sqrt{3}$ terms cancel in the average. What a cube adds to a sphere at this
 527 order is therefore a pure cubic anisotropy. It splits the deviatoric channel
 528 about the sphere's value, weighted 3 : 2 by the dimensions of T_{2g} and E_g ,
 529 and the diagonal shear is the more depolarised. At the background of §7

530 $(\nu = 7/32)$, $2\mu S_{\text{off}} = 0.4314$, $2\mu S_{\text{diag}} = 0.5929$ and the sphere's value is
531 0.4960.

532 *Where Eshelby enters.* Equations (29)–(31) follow from the moment integrals
533 and the group theory alone. No ellipsoidal result is used, and no δ term is
534 added by hand. What corresponds to Eshelby is the *truncation*. Keeping
535 ∂u and discarding $\partial\partial u$ is exactly the uniform-internal-strain ansatz, and by
536 the uniformity theorem that ansatz is exact if and only if the inclusion is an
537 ellipsoid. For a sphere the construction is therefore exact in the static limit,
538 and for a cube it is an approximation. The hierarchy (15)–(17) states that
539 approximation as the omitted blocks, rather than as an unquantified shape
540 correction. A polynomial expansion about the centre cannot represent the
541 edge and corner concentrations of an isolated cube's internal field at any finite
542 degree; Table 4 measures what each degree leaves. A voxel in a uniform layer
543 has no such concentrations: its neighbours have the same properties, so its
544 internal field is the continuum's, which is smooth. That is why a low-degree
545 internal field suffices in a lattice (§§7–8) although it is only an approximation
546 for an isolated cube.

547 *Against the exact sphere.* These statements are tested against the exact elas-
548 tic Mie solution for a sphere, with no Eshelby formula in between. The
549 concentration of Mie order n is $E_n = a_n/a_n^{\text{Born}}$ (orders 0, 1, 2 are the dilata-
550 tional, density and shear channels), extrapolated to $ka \rightarrow 0$. The formulation
551 is applied to a ball of radius a , with its moment $\mathbf{M}$ evaluated by the surface
552 route of Eq. (25) on $|\mathbf{r}| = a$. Its dilatational and shear channels equal Mie's
553 to 2×10^{-10} , and the density channel is 1. The cube's closed forms meet
554 the same numbers. Its dilatational channel is Mie's E_0 to 2×10^{-10} , and its

isotropic shear average, Eq. (32), is the depolarisation read from Mie's E_2 to 10^{-8} . The same moment taken as a principal value, over the ball with a small sphere excluded around the origin, has zero depolarisation in every channel. The whole static response of the sphere is the δ term.

At finite frequency the closure uses the dynamic ball moments, which are in closed form. It departs from Mie at second order in ka and first order in the contrast, and the departure is closed-form too: the ball's moments are elementary, and Mie's low-frequency series with the contrast kept exact gives E_n to relative order $(ka)^3$. To first order in the contrast,

$$\frac{E_n^{\text{closure}}}{E_n^{\text{Mie}}} - 1 = \begin{cases} -\frac{(k_P a)^2}{10} \frac{\Delta\lambda}{\lambda + 2\mu}, & n = 0, \\ \frac{2(k_S a)^2 + (k_P a)^2}{30} \frac{\Delta\rho}{\rho}, & n = 1, \\ -\frac{3(k_S a)^2 + 2(k_P a)^2 \beta^2 / \alpha^2}{75} \frac{\Delta\mu}{\mu}, & n = 2, \end{cases} \quad (33)$$

with no $O(ka)$ term and no $O((ka)^3)$ term. The imaginary $(ka)^3$ terms, the radiation damping, agree exactly in all three channels. So the closure's only error at this order is the real $(ka)^2$ term, the uniform-field ansatz's own. That is the order the hierarchy assigns to the omitted second gradients (§6.3), and it has the same origin, the variation of the field across the scatterer, as the collocation error of §7. Eq. (33) is derived symbolically (`SphereClosureDynamic.wl`). It agrees with the coefficient extracted from the numerical Mie solve and an independent implementation of the closure to 5×10^{-4} , which is the size of the next order in the contrast.

573 *6.4. What each block of the hierarchy buys*

574 A hierarchy invites the question of what its next block is worth. The
 575 answer has a simple form, and it is not the one a count of unknowns suggests:
 576 the blocks help in pairs.

577 *The experiment.* The layer of §7 at normal incidence is cut into n cells, and
 578 in each the hierarchy of degree ℓ is imposed: the displacement is its Taylor
 579 polynomial of degree ℓ about the centre, the strain the derivative of that, and
 580 the equation and its first ℓ derivatives hold at the centre, with the δ terms
 581 kept. In one dimension the moments are elementary and the exact layer is
 582 known, so the order of each truncation can be measured for a density contrast,
 583 a modulus contrast and both. The schemes are computed symbolically to 60
 584 digits for $\ell = 1$ to 6 and, to $\ell = 3$, independently in double precision.

585 *The prediction.* A cell radiates the integral of its source against the kernel.
 586 If the source is held to a polynomial of degree d about the centre, the first
 587 power it omits is $d + 1$. When $d + 1$ is odd, that term integrates to zero
 588 against the even part of the kernel and the error is two orders smaller; when
 589 $d + 1$ is even it does not. The density source is the displacement, of degree ℓ ;
 590 the stiffness source is the strain, of degree $\ell - 1$. So a density contrast gains
 591 an order pair at every even ℓ , and a stiffness contrast at every odd ℓ . This is
 592 the parity of §6.3, read as an order of convergence.

593 *The result.* Table 2: all thirty-six orders are as predicted. The second gradi-
 594 ents raise the density channel to fourth order and leave the stiffness channel
 595 at second, with a constant three to four times smaller in reflection; the third
 596 gradients bring the stiffness channel to fourth order; and so on in pairs. With

Table 2: The order of convergence of the hierarchy of degree ℓ on the layer, measured from 16 to 32 cells in reflection and in transmission at $\omega = 300 \text{ rad s}^{-1}$, and its reflection error on 16 cells with both contrasts. Every measured order equals the predicted one to the digits shown.

	$\ell = 1$	2	3	4	5	6
density contrast only	2	4	4	6	6	8
modulus contrast only	2	2	4	4	6	6
both	2	2	4	4	6	6
error, both, $n = 16$	5.4×10^{-6}	1.4×10^{-6}	5.8×10^{-12}	9.5×10^{-13}	2.6×10^{-18}	3.1×10^{-19}

both contrasts, fourth order therefore needs $\ell = 3$ and sixth order $\ell = 5$. The truncation of Eqs. (15)–(17) at second gradients is one block short of the next order for a stiffness contrast, which is why it improves the uniform-strain closure by a constant and not by an order.

The error in closed form. At first order in the contrast a cell whose source is held to its Taylor polynomial of degree d returns the exact term times $1 + E_d$, where, with h the half-width,

$$E_d = \begin{cases} (-1)^{(d-1)/2} \frac{(kh)^{d+1}}{(d+2)!} & d \text{ odd, } R \text{ and } T, \\ (-1)^{d/2} \frac{(kh)^{d+2}}{(d+2)!} & d \text{ even, } R, \\ (-1)^{d/2+1} \frac{(d+1)(kh)^{d+2}}{(d+3)!} & d \text{ even, } T, \end{cases} \quad (34)$$

R and T standing for reflection and transmission, with $d = \ell$ for the density term and $d = \ell - 1$ for the stiffness term. It follows from the series of $\int_{-1}^1 P_d(ikhx) e^{\mp ikhx} dx$ with P_d the Taylor polynomial of e^{ikhx} , and it is

confirmed symbolically for $d = 0$ to 6 and against the scheme on one cell, at a contrast of 10^{-20} and $kh = 0.03$, to 0.5% in all twelve cases. For $d = 0$ it is $(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission: the constants of the collocation scheme, Eq. (37), which are thus the first rung of a ladder (`ContinuumLimit_GradientHierarchy.wl`, `scripts/measure_layer_taylor_hierarchy.py`).

6.5. The single site against a sphere and a cube

The hierarchy is a statement about one scatterer, so it is tested first on one scatterer. The assembly of Eq. (21), with the full tensor moments in their dynamic form (the series of e^{ikr}/r taken to r^{11}), is the same for any centrosymmetric body; only the scalar moments change.

A sphere. For a ball the scalar moments are elementary, by the surface route of Eq. (25) on $|\mathbf{r}| = a$, and the exact answer is known. A homogeneous sphere of radius $a = 10$ m with the contrast of §7 is illuminated by a P plane wave; the hierarchy's polynomial source is radiated to nine scattering angles between 11° and 169° , and the error is the largest component of the difference from the exact displacement [47, 48] over the largest exact component. The prediction, from the pairs of §6.4 with kh replaced by ka , is an error of $O((ka)^2)$ for $\ell = 1$ and 2 and of $O((ka)^4)$ for $\ell = 3$, and $O((ka)^4)$ already at $\ell = 2$ for a density contrast alone. Table 3 bears the prediction out. At $k_S a = 0.4$ the error of the uniform-strain closure is 1.4%, that of the second-gradient system 0.32%, and that of the third-gradient system 0.0046%: three hundred times below the closure. The systems are well conditioned, the sixty equations of $\ell = 3$ having a condition number of 5.9 (`scripts/measure_`

Table 3: One homogeneous sphere by the hierarchy of degree ℓ : relative far-field error against exact scattering, and the order between the two lowest frequencies. All nine orders are as predicted.

contrast	ℓ (unknowns)	$k_{Sa} = 0.4$	0.2	0.1	order	predicted
density only	1 (12)	6.4×10^{-3}	1.6×10^{-3}	3.9×10^{-4}	2.0	2
	2 (30)	4.8×10^{-5}	2.6×10^{-6}	1.8×10^{-7}	3.9	4
	3 (60)	1.5×10^{-5}	9.5×10^{-7}	6.9×10^{-8}	3.8	4
moduli only	1	1.9×10^{-2}	5.0×10^{-3}	1.1×10^{-3}	2.3	2
	2	5.4×10^{-3}	1.4×10^{-3}	3.4×10^{-4}	2.0	2
	3	6.2×10^{-5}	4.1×10^{-6}	2.0×10^{-7}	4.3	4
both	1	1.4×10^{-2}	3.1×10^{-3}	7.2×10^{-4}	2.1	2
	2	3.2×10^{-3}	6.9×10^{-4}	1.8×10^{-4}	1.9	2
	3	4.6×10^{-5}	2.5×10^{-6}	1.4×10^{-7}	4.2	4

631 `ball_gradient_hierarchy.py`).

632 *A cube.* The same assembly with the cube's moments gives the single site
633 of a voxel. No exact solution exists for a cube, so the reference is the same
634 cube, of half-width $h = 10$ m, solved by the Legendre voxels of §8 on refined
635 grids, a cube being the one body such a grid represents exactly. Solutions
636 with first and second moments on 2^3 and 3^3 cells differ among themselves
637 by at most 2×10^{-5} , which is the accuracy of the reference. Table 4 shows
638 what distinguishes a cube from a sphere. Each truncation has a floor that
639 does not fall with the frequency, because the internal field of an isolated
640 cube is not a polynomial even statically: it concentrates at the edges and
641 corners. The floor is 2.7×10^{-3} for the uniform-strain closure and nearly the

Table 4: One homogeneous cube of half-width h by the hierarchy of degree ℓ : relative far-field error against the converged reference.

ℓ (unknowns)	$k_S h = 0.4$	0.2	0.1	0.05
1 (12)	2.1×10^{-2}	5.4×10^{-3}	2.8×10^{-3}	2.7×10^{-3}
2 (30)	3.2×10^{-3}	1.6×10^{-3}	1.9×10^{-3}	2.4×10^{-3}
3 (60)	3.5×10^{-4}	3.4×10^{-4}	2.9×10^{-4}	3.1×10^{-4}

642 same with second gradients, as §6.3 requires, the static stress dipole being
 643 fixed by the first-gradient block and coupled next to the third. The third-
 644 gradient block lowers it ninefold, to 3×10^{-4} . Above the floor the error
 645 falls with the frequency as on the sphere: at $k_S h = 0.4$ the third-gradient
 646 system is sixty times more accurate than the closure. The dip of the second-
 647 gradient row at $k_S h = 0.2$ is a cancellation between its static and dynamic
 648 terms, which have opposite signs. The condition numbers lie between 11 and
 649 13 (`scripts/measure_cube_gradient_hierarchy.py`). These results come
 650 from one implementation, checked against the exact sphere and against the
 651 converged cube; the moments it uses are those confirmed by two routes in
 652 §6.2.

653 6.6. The collocation closure

654 The collocation closure takes the internal field of a voxel as uniform, the
 655 truncation of §6.3 for the density and stiffness channels together. The voxel's
 656 own field at its centre is then $\mathbf{S}_{\text{self}} \Delta e$ for internal state e , where $\mathbf{S}_{\text{self}}$ is the
 657 nine-component form of the moments $\mathbf{G}$ and $\mathbf{M}$. By parity it is block-diagonal.

658 So $e = \psi + \mathbf{S}_{\text{self}} \Delta e$ and

$$\mathbf{T} = d^3 \Delta (\mathbf{I} - \mathbf{S}_{\text{self}} \Delta)^{-1}. \quad (35)$$

659 With the kernel (14) the self term cancels. Substituting $\psi = (\mathbf{I} - \mathbf{S}_{\text{self}} \Delta)e$
 660 into Eq. (2),

$$e_i = \psi_i^0 + d^3 \sum_j \mathbf{K}_{\text{all}}(i, j) \Delta e_j, \quad \mathbf{K}_{\text{all}}(i, i) = \mathbf{P}_0, \quad \mathbf{K}_{\text{all}}(i, j) = \mathbf{S}_{\text{avg}}(i - j), \quad (36)$$

661 an identity checked on the matrices to 4×10^{-16} . Every block of $\mathbf{K}_{\text{all}}$ is an
 662 *exact* cell integral of the one-dimensional continuum kernel (Eqs. (12), (13)),
 663 so Eq. (36) is precisely the collocation of the continuum Lippmann–Schwinger
 664 equation with a strain piecewise constant over voxels. It does not contain
 665 $\mathbf{S}_{\text{self}}$ at all.

666 The package’s cube T -matrix is this closure statically: both blocks agree
 667 with Eq. (35) to 2×10^{-5} at $k_S h = 0.01$, the difference falling as $(kh)^2$
 668 (`ContinuumLimit_CubeT.wl`). It departs dynamically by about $0.25(kh)^2$
 669 through its far-field form-factor corrections. Its self integrals are exact in the
 670 static and radiation parts but omit the real even-power terms of the Green’s
 671 tensor’s expansion ($k^2 r, r^3, \dots$), an error of 0.4–0.6 $(kh)^2$ against the analytic
 672 self term, whose series agrees with the closed-form elastodynamic tensor to
 673 10^{-14} . By Eq. (36) that error cannot enter the closure scheme.

674 7. Convergence, and the closed-form error

675 *The experiment.* Every convergence test of this paper is the same experiment,
 676 varied in one respect at a time, and it is set out here once.

- 677 1. *The medium.* The background has $\alpha = 5000$ and $\beta = 3000 \text{ m s}^{-1}$ and
678 $\rho = 2500 \text{ kg m}^{-3}$. The layer occupies $0 < z < D$ with $D = 2 \text{ m}$,
679 and differs from the background by $\Delta\lambda = 2 \text{ GPa}$, $\Delta\mu = 1 \text{ GPa}$ and
680 $\Delta\rho = 100 \text{ kg m}^{-3}$: 6.4% in the P modulus, 4.4% in the shear modulus
681 and 4% in the density.
- 682 2. *The source.* A plane force at $z_s = -12 \text{ m}$, above the layer, radiates a
683 plane wave towards it. The incident field in the layer is then known in
684 closed form, and it is smooth there because the source is outside.
- 685 3. *What is measured.* The scattered displacement, total less incident, at
686 two observers outside the layer: at $z = -1 \text{ m}$ above it, where it is the
687 reflected wave, and at $z = 4 \text{ m}$ below it, where it is the change the layer
688 makes to the transmitted wave. Outside the layer the scattered field is
689 a single outgoing wave, so one observer on each side holds everything;
690 the two are reported separately, as reflection / transmission.
- 691 4. *The exact answer.* The reflection and transmission of the layer from
692 continuity of displacement and traction at its two faces (§3); for a
693 layer whose properties vary with depth, from the differential equation
694 integrated to 30 digits.
- 695 5. *The error.* $e = |u_{\text{scheme}} - u_{\text{exact}}|/|u_{\text{exact}}|$ at each observer: relative to the
696 scattered field, not to the total, so that an error of 10^{-6} is a millionth
697 of what the layer scatters.
- 698 6. *The refinement.* The layer is cut into n planes of voxels of side $d =$
699 D/n , with n from 1 to 32, at $\omega = 60, 300$ and 600 rad s^{-1} , for which
700 $k_P D = 0.024, 0.12$ and 0.24 and $k_S D = 0.04, 0.2$ and 0.4 . Three
701 frequencies show that an order is not an accident of one.

702 7. *The arithmetic.* The errors of the higher-degree voxels fall below the
703 round-off of double precision, so the symbolic implementation computes
704 them to 40 digits; the independent numerical implementation, in double
705 precision, agrees with it down to its own floor of about 10^{-13} .

706 The later sections change one item each: the incident wave to SV or SH and
707 its angle (§8), the medium to a stratified or a graded one and the background
708 to a stratified one (§9), and the body to a sphere (§10).

709 *Second order.* For a layer $D = 2\text{ m}$ with the contrast $\Delta\lambda = 2\text{ GPa}$, $\Delta\mu =$
710 1 GPa , $\Delta\rho = 100\text{ kg m}^{-3}$ in a background $\alpha = 5000$, $\beta = 3000\text{ m s}^{-1}$, $\rho =$
711 2500 kg m^{-3} , illuminated by a P plane force, the reflected displacement of
712 the chain converges to the continuous layer's at order 2.000 for $n = 4 \rightarrow$
713 16 at $\omega = 60, 300$ and 600 rad s^{-1} , with the collocation (36) and with the
714 package *T*-matrix alike (`ContinuumLimit_Chain.wl`). The error depends on
715 $k_S h$ alone. Throughout, an order quoted for a pair of grids is the apparent
716 order $\ln(e_1/e_2)/\ln(n_2/n_1)$ between them, with e the relative error against
717 the exact solution and n the refinement; the orders in the legend of Fig. 6
718 are least-squares slopes of $\ln e$ against $\ln n$ over the grids marked as fitted,
719 with their standard errors. The two agree where a sequence is asymptotic
720 and differ where it is not, which the apparent orders show.

721 *The leading error in closed form.* In Eq. (36) the kernel is integrated exactly;
722 the only approximation is the internal field's variation across a voxel. At first
723 order in the contrast the discrete scattered field is therefore a midpoint rule
724 applied to the continuum's Born integral, with the kernel's cell integral exact.
725 For reflection the integrand carries the two-way phase e^{2ikz} ; for transmission

726 it is flat. With k the background wavenumber of the wave concerned (k_P for
 727 an incident P wave, k_S for an incident S wave), R the reflection coefficient
 728 and T^{sc} the scattered part of the transmitted field, for *every* n

$$\frac{R_{\text{disc}}}{R} = \frac{\text{sinc}(kh)}{\text{sinc}(2kh)} = 1 + \frac{(kd)^2}{8} + O(d^4), \quad \frac{T_{\text{disc}}^{\text{sc}}}{T^{\text{sc}}} = \text{sinc}(kh) = 1 - \frac{(kd)^2}{24} + O(d^4), \quad (37)$$

729 confirmed for $n = 1$ to 16 to 10^{-4} of the error itself (`ContinuumLimit_`
 730 `ErrorConstant.wl`); for an incident S wave, with $k = k_S$, the same forms
 731 and those of Eq. (40) hold to 4×10^{-4} (`ContinuumLimit_IncidentS.wl`, §8).
 732 At the full contrast the reflection constant is unchanged (0.124 against 1/8);
 733 the transmission constant grows to -0.092 .

734 *Why a uniform-field single site stops at second order.* The two constants have
 735 opposite signs, so no scalar correction of $\mathbf{T}$, no single $O((kh)^2)$ factor, can
 736 cancel both. The package T -matrix illustrates it: its form-factor corrections
 737 reduce the reflection error by a factor 4.45 and *increase* the transmission
 738 error by 1.7, at every n . The missing term in Eq. (37) is the first moment
 739 of the internal field over each voxel, which radiates through the derivative of
 740 the kernel. A fourth-order scheme must carry it: a single site with a gradient
 741 channel, and a kernel whose cell integrals include the kernel's derivative.

742 8. Fourth order and beyond: Legendre moments of the voxel field

743 Let each voxel carry the mean *and* the first moment of its internal field.
 744 On voxel j , with local coordinate $s \in [-h, h]$ and Legendre functions $\phi_b(s) =$
 745 $P_b(s/h)$ ($\phi_0 = 1$, $\phi_1 = s/h$, $\phi_2 = (3s^2/h^2 - 1)/2$, ...), the field is $w =$
 746 $\sum_b c_{b,j} \phi_b$, $w = (u, \varepsilon)$ the displacement and strain of the P channel, and the

747 polarisation is $q = (\omega^2 \Delta \rho u, \Delta M \varepsilon)$ with $\Delta M = \Delta \lambda + 2\Delta \mu$. Testing the
748 continuum equation

$$w(z) = w^0(z) + \int \mathbf{K}(z-z') q(z') dz', \quad \mathbf{K} = \begin{pmatrix} g & g' \\ g' & g'' \end{pmatrix}, \quad g'' = -k^2 g - \delta(z)/M_P, \quad (38)$$

749 with the same functions on every voxel gives

$$\left(\int_i \phi_a^2 dz \right) c_{a,i} = \int_i \phi_a w^0 dz + \sum_{j,b} \left[\iint_j \phi_a(z) \mathbf{K}(z-z') \phi_b(z') dz dz' \right] \Delta_1 c_{b,j}, \quad (39)$$

750 $\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$. Between voxels the double integral is separable, a
751 product of Legendre moments of $e^{\pm i k z}$; within a voxel it is split at $z' = z$
752 and the local term contributes $-\int_i \phi_a \phi_b dz / M_P$. Every entry is closed-form.
753 Keeping only ϕ_0 recovers a second-order scheme; keeping ϕ_0 and ϕ_1 con-
754 verges at fourth order in both reflection and transmission (`ContinuumLimit_`
`FourthOrder.wl`; Table 5). A single plane of first-moment voxels represents

Table 5: Relative error of the scattered displacement of the layer, reflection / transmission, at $\omega = 300 \text{ rad s}^{-1}$, against the number of voxel planes n : collocation, the mean-only voxel and the first-moment voxel of Eq. (39). The order is 4.00 at $\omega = 60$ and 600 rad s^{-1} too.

n	collocation	mean only	mean and first moment
1	$1.8 \times 10^{-3} / 1.3 \times 10^{-3}$	$1.2 \times 10^{-3} / 1.8 \times 10^{-3}$	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$
4	$1.1 \times 10^{-4} / 8.3 \times 10^{-5}$	$7.5 \times 10^{-5} / 1.1 \times 10^{-4}$	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$
16	$7.0 \times 10^{-6} / 5.2 \times 10^{-6}$	$4.7 \times 10^{-6} / 6.9 \times 10^{-6}$	$4.2 \times 10^{-12} / 4.2 \times 10^{-12}$
order	2 / 2	2 / 2	4 / 4

755

756 the 2 m layer 6500 times more accurately than a single plane of uniform ones.

757 The collocation column is the scheme of §6 and reproduces the chain of the
758 full 9×9 system to all printed digits.

759 *Higher moments.* Carrying the Legendre moments to degree p changes noth-
760 ing in the construction: the moments and the same-cell double integrals of
761 $P_b(s/h)$ against exponentials are again closed-form, and for $p \leq 1$ the general-
762 degree solver reproduces the chain above exactly. The second-moment voxel
763 ($p = 2$) converges at sixth order and the third-moment voxel ($p = 3$) at
764 eighth, in reflection and transmission (Table 6; `ContinuumLimit_SecondMoment`.
765 `w1`); the second-moment voxel’s order is 6.00 at $\omega = 60, 300$ and 600 rad s^{-1} .
The errors fall to 10^{-22} , so they are computed in 40-digit arithmetic.

Table 6: Relative error of the scattered displacement of the layer, reflection / transmission,
at $\omega = 300 \text{ rad s}^{-1}$, for voxels carrying Legendre moments to degree p . The order is the
mean over the halvings $n = 2 \rightarrow 4 \rightarrow 8$.

n	first moment ($p = 1$)	second moment ($p = 2$)	third moment ($p = 3$)
1	$2.8 \times 10^{-7} / 2.8 \times 10^{-7}$	$2.8 \times 10^{-11} / 4.2 \times 10^{-11}$	$1.5 \times 10^{-15} / 1.5 \times 10^{-15}$
2	$1.7 \times 10^{-8} / 1.7 \times 10^{-8}$	$4.4 \times 10^{-13} / 6.5 \times 10^{-13}$	$6.0 \times 10^{-18} / 6.0 \times 10^{-18}$
4	$1.1 \times 10^{-9} / 1.1 \times 10^{-9}$	$6.9 \times 10^{-15} / 1.0 \times 10^{-14}$	$2.4 \times 10^{-20} / 2.4 \times 10^{-20}$
8	$6.7 \times 10^{-11} / 6.7 \times 10^{-11}$	$1.1 \times 10^{-16} / 1.6 \times 10^{-16}$	$9.2 \times 10^{-23} / 9.2 \times 10^{-23}$
order	4 / 4	6 / 6	8 / 8

766

767 *The leading error in closed form.* Without scattering, the Galerkin solution
768 is the L^2 projection Π of the incident field onto each voxel’s basis, and the
769 kernel is integrated over each voxel exactly; at first order in the contrast the
770 error is therefore the kernel’s cell integral applied to $\Pi w^0 - w^0$. The field
771 varies as e^{iks} across a voxel and the kernel toward an outside observer as e^{iqs} ,

772 $q = +k$ for reflection and $q = -k$ for transmission, so for every scheme and
 773 every n

$$\frac{\text{discrete}}{\text{exact}} = \frac{\int_{-h}^h e^{iqs} \Pi[e^{iks}] ds}{\int_{-h}^h e^{i(q+k)s} ds}, \quad (40)$$

774 with Π the centre value for collocation, the cell mean for the mean-only voxel
 775 and the projection onto $\{P_0, \dots, P_p\}$ for the voxel of degree p (`ContinuumLimit_`
 776 `FourthOrderError.wl`, `ContinuumLimit_SecondMoment.wl`). The results
 are in Table 7. The chain of Eq. (39) reproduces the forms for collocation

Table 7: The leading error of the discrete layer in closed form, Eq. (40): discrete over exact scattered field at first order in the contrast, for every number of voxel planes n .

	reflection	transmission
collocation	$1 + (kd)^2/8 + O(d^4)$	$1 - (kd)^2/24 + O(d^4)$
mean only	$1 + (kd)^2/12 + O(d^4)$	$1 - (kd)^2/12 + O(d^4)$
first moment	$1 - (kd)^4/720 + O(d^6)$	$1 - (kd)^4/720 + O(d^6)$
second moment	$1 + (kd)^6/100800 + O(d^8)$	$1 - (kd)^6/100800 + O(d^8)$
third moment	$1 - (kd)^8/25401600 + O(d^{10})$	$1 - (kd)^8/25401600 + O(d^{10})$
degree p	$1 + (-1)^p c_p (kd)^{2p+2}$	$1 - c_p (kd)^{2p+2}$

777
 778 and degrees 0 and 1 at $n = 1$ to 8 to 10^{-4} of the error, the remainder being
 779 the second order in the contrast, and those for degrees 2 and 3 to 4×10^{-7}
 780 at a contrast reduced by 10^{-6} . The constant for degree p follows from the
 781 Legendre expansion $e^{iks} = \sum_j i^j (2j+1) j_j(kh) P_j(s/h)$. The projection drops
 782 the terms $j > p$, led by $j = p+1$, whose coefficient is $O((kh)^{p+1})$; its moment
 783 against e^{iqs} is $2h i^{p+1} j_{p+1}(qh) = O((qh)^{p+1})$. With $j_n(x) \approx x^n/(2n+1)!!$ and

784 $h = d/2,$

$$c_p = \frac{2p+3}{4^{p+1}[(2p+3)!!]^2} = \frac{[(p+1)!]^2}{(2p+2)!(2p+3)!}, \quad c_0 = \frac{1}{12}, c_1 = \frac{1}{720}, c_2 = \frac{1}{100800},$$

(41)

785 and the leading error carries $(q/k)^{p+1}$: the same sign in reflection and trans-
 786 mission for odd p , opposite signs for even p . The first-moment voxel's leading
 787 error therefore does not depend on the direction in which the field is observed.
 788 Eq. (41) is checked symbolically for $p = 0$ to 5 (every lower power of kd van-
 789 ishes) in Mathematica and, independently, in sympy.

790 *One plane of voxels and the thin-layer series.* Refining at fixed thickness is
 791 one limit; thinning a single plane is the other. For one plane of voxels ($n = 1$,
 792 $d = D$) the series in D of the reflection coefficient, computed exactly at the
 793 full contrast, coincides with that of the exact layer, Eq. (7), term by term
 794 through D^{2p+2} and first differs at D^{2p+3} , for $p = 0$ to 3 and for a layer both
 795 stiffer and softer than its surroundings (`ContinuumLimit_SecondMoment.`
 796 `w1`). Each further Legendre moment adds two terms of the thin-layer series.

797 *In three dimensions.* The three-dimensional first-moment voxel carries the
 798 nine-component state with Legendre weights $\{1, x/h, y/h, z/h\}$, 36 unknowns
 799 in all, tested with the same functions and coupled by cell-to-cell double in-
 800 tegrals of the Green's tensor. For a laterally uniform layer at normal in-
 801 cidence it reduces *exactly* to Eq. (39) (`ContinuumLimit_Reduction3D.w1`).
 802 The lateral first moments are never excited: the lattice and the incident
 803 field are symmetric under reflection about every voxel's centre planes, and
 804 $x/h, y/h$ are odd there. The mean and the z -moment carry weights that
 805 are constant laterally, so their lateral form factor $\text{sinc}(g_x h) \text{sinc}(g_y h)$ van-

ishes on the reciprocal lattice and the plane sum of the double integrals is the plate-to-plate integral in z alone. Two facts mark where genuinely three-dimensional structure begins. The lateral first moment's form factor, $F_1(g) = i(gh \cos gh - \sin gh)/(gh)^2$, does not vanish on the reciprocal lattice ($F_1 = i(-1)^p/p\pi$), so the gradient channels couple through the evanescent orders. And at oblique incidence the mean's own zeros shift, $\text{sinc}((2\pi p/d + k_x)h) = (-1)^p k_x h/p\pi + O(k_x^2 h^2)$, so the tiling identity holds only to $O(k_{\parallel}h)$.

Oblique incidence. The scheme is built spectrally. By Poisson summation the Bloch coupling between planes i and j at lateral wavenumber $\mathbf{k}_{\parallel}$ is

$$C_{ab}(i, j) = \frac{1}{d^2} \sum_{\mathbf{g}} L_a(\boldsymbol{\kappa}) L_b(-\boldsymbol{\kappa}) \iint \phi_a(z) \hat{\mathbf{G}}(\boldsymbol{\kappa}; z-z') \phi_b(z') dz dz', \quad \boldsymbol{\kappa} = \mathbf{k}_{\parallel} + \mathbf{g}, \quad (42)$$

with L_a the lateral Legendre factors. The whole-space tensor decays as $1/k_z^2$ and the nine-component state adds at most two powers of k_z , so each entry of $\hat{\mathbf{G}}(\boldsymbol{\kappa}; z)$ is a $\delta(z)$ of constant weight plus, per mode, $[i\alpha/2\gamma + i\beta \text{sgn}(z)/2] e^{i\gamma|z|}$, $\gamma^2 = k_W^2 - \kappa^2$: every z integral is closed-form. Double averaging makes the lattice sum converge without Ewald splitting, to three digits at $|p|, |q| \leq 8$. Against the exact layer, for a P wave incident at 20° and 40° (`ContinuumLimit_Oblique.wl`; Table 8). The mean-only voxel is second order and the first-moment voxel fourth order in both the reflected P and the converted S wave: the breakdown of the tiling identity at $O(k_{\parallel}h)$ and the evanescent coupling of the gradient channels do not cost an order. At $\mathbf{k}_{\parallel} = 0$ the construction reproduces Eq. (39) to all printed digits.

Table 8: Relative error of the reflected P (R_{PP}) and converted S (R_{PS}) waves for a P wave incident at 20° , with the mean-only and first-moment voxels of Eq. (42). At 40° the orders are again 2 and 4.

	mean only		mean and first moments	
n	R_{PP}	R_{PS}	R_{PP}	R_{PS}
1	9.3×10^{-4}	1.7×10^{-3}	7.2×10^{-8}	4.1×10^{-7}
2	2.3×10^{-4}	4.2×10^{-4}	4.5×10^{-9}	2.5×10^{-8}
8	1.5×10^{-5}	2.7×10^{-5}	1.7×10^{-11}	9.9×10^{-11}
order	2	2	4	4

827 *Incident S.* Nothing in the construction is specific to P. For a normally in-
828 cident S wave the layer is the one-dimensional problem of Eq. (39) with
829 $M_P \rightarrow \mu$, $k_P \rightarrow k_S$ and $\Delta M \rightarrow \Delta\mu$; the collocation and mean-only schemes
830 converge at order 2.00 and the first-moment voxel at 4.00, in reflection and
831 transmission, and the full 9×9 construction with an incident SH wave re-
832 produces that one-dimensional result to all printed digits. At oblique inci-
833 dence, against the exact layer (`ContinuumLimit_IncidentS.wl`), the orders
are those of Table 9. The orders, and with $k = k_S$ the closed-form constants

Table 9: Observed order of convergence ($n = 2 \rightarrow 8$, $\omega = 300 \text{ rad s}^{-1}$) for incident SV and SH waves, with the mean-only and first-moment voxels. The SV angles lie below the critical angle $\arcsin(\beta/\alpha) = 36.9^\circ$.

	mean only	mean and first moments
SV at 20° : R_{SS} / R_{SP}	2.00 / 2.00	4.00 / 3.99
SV at 30° : R_{SS} / R_{SP}	1.98 / 2.00	3.97 / 4.00
SH at 20° and 40° : R_{SH}	2.00	4.00

834 of Eqs. (37) and (40), are those of the incident P wave.

835 *Degree two at oblique incidence.* In three dimensions the voxel of degree 2
836 carries, on each of the nine components of its state, the ten Legendre products
837 of total degree at most two in (z, x, y) . In the construction of Eq. (42) it
838 converges at sixth order against the exact layer for incident P, SV and SH
839 waves (Table 10; `scripts/measure_oblique_degree.py`). The test is run at
840 $\omega = 3000 \text{ rad s}^{-1}$ ($k_P D = 1.2$, $k_S D = 2$), ten times the frequency of the other
841 tables, because at 300 rad s^{-1} a sixth-order error on these grids lies below
842 the round-off of double precision. The exact reflection coefficient is that of
843 Kennett’s recursion, an independent algorithm, and the same computation
844 with the bases of degree 0 and 1 returns orders 2 and 4 in all five cases.
845 At normal incidence the error on four planes, 6.95×10^{-9} , is the $c_2(kd)^6 =$
846 7.2×10^{-9} of Eq. (41) to 4%. Doubling the number of lattice orders retained
847 in Eq. (42) changes the errors by less than 1%. Unlike the degrees 0 and 1,
848 this degree is implemented once at oblique incidence; its check is the exact
849 solution.

850 The spectral construction is implemented twice. The second implementa-
851 tion (`scripts/crosscheck_first_moment_voxel.py`) shares only the defi-
852 nition of the scheme: it takes the depth dependence of the transformed kernel
853 from an independent residue construction, obtains the local weight as a nu-
854 merical large- k_z limit of the inverted Christoffel matrix, evaluates every dou-
855 ble integral by quadrature and separates P from S by sampling the upgoing
856 field at two depths. The raw reflected displacements of the two agree to $9 \times$
857 10^{-16} in all 36 cases compared: incident P, SV and SH, normal incidence and
858 20° , $n = 1, 2, 4$, with and without the first moments. The one-dimensional

Table 10: The voxel of degree 2 (ten Legendre products per state component) at $\omega = 3000 \text{ rad s}^{-1}$: relative error of the same-type reflection coefficient against the exact layer on n planes, and the order over the last halving, with the orders of the degrees 0 and 1 from the same computation.

incident wave	error, $p = 2$			order	orders
	$n = 1$	$n = 2$	$n = 4$	$p = 2$	$p = 0$ / $p = 1$
P, normal: R_{PP}	3.4×10^{-5}	4.6×10^{-7}	7.0×10^{-9}	6.05	2.04 / 4.05
P at 20° : R_{PP}	7.3×10^{-6}	1.0×10^{-7}	1.5×10^{-9}	6.04	2.04 / 4.06
SV at 20° : R_{SS}	2.2×10^{-4}	2.3×10^{-6}	3.4×10^{-8}	6.11	2.10 / 4.14
SH at 20° : R_{SH}	2.3×10^{-4}	2.5×10^{-6}	3.6×10^{-8}	6.11	2.10 / 4.14
SH at 40° : R_{SH}	3.0×10^{-4}	3.6×10^{-6}	5.2×10^{-8}	6.09	2.12 / 4.05

859 scheme of any degree is also implemented twice. The second implementation
860 (`scripts/crosscheck_second_moment_voxel.py`) evaluates every integral
861 by Gauss quadrature, split at the receiver within a voxel, and the exact layer
862 from its own interface system, in double precision. Its errors agree with the
863 40-digit ones within its round-off floor (3×10^{-14} in reflection, 1.6×10^{-13} in
864 transmission); at $\omega = 1200 \text{ rad s}^{-1}$ it measures orders 4.02 / 3.99 and 6.02 /
865 6.00; for one plane as the layer thins its differences from the exact layer fall
866 as $D^{3.00}$, $D^{5.00}$ and $D^{7.00}$ for $p = 0, 1$ and 2 ; and it rederives c_p for $p \leq 5$.
867 Figure 6(a,b) shows the convergence, with the orders fitted to it.

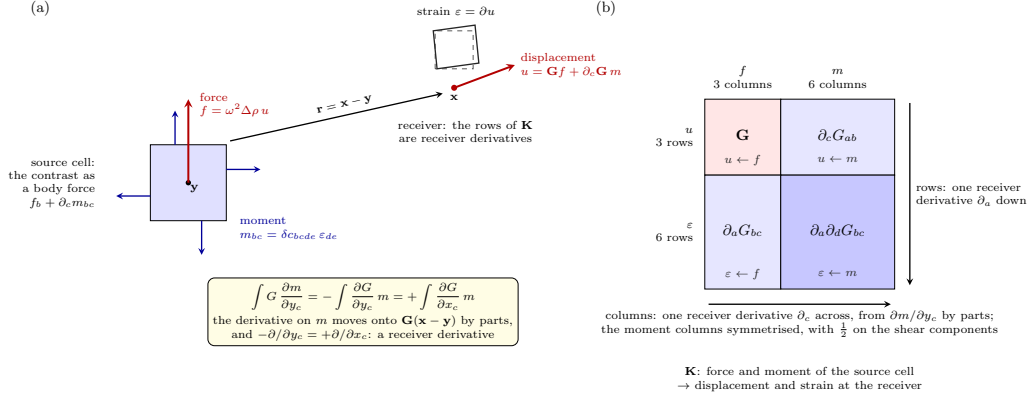


Figure 1: Where the kernel comes from. (a) A source cell at $\mathbf{y}$ acts on the background through the body force of Eq. (3): a force f from its density contrast, and the divergence of its contrast stress m , the moment, from its stiffness contrast (arrows on the faces). At the receiver $\mathbf{x}$ the state is the displacement u and the strain ε . The moment's derivative moves onto $\mathbf{G}(\mathbf{x} - \mathbf{y})$ by parts, Eq. (4), and becomes a receiver derivative with a plus sign. (b) The block structure of the 9×9 kernel $\mathbf{K}$ that results: the force columns are $\mathbf{G}$, the moment columns carry one receiver derivative across, the strain rows one receiver derivative down, and the strain-from-moment block two.

Alt text: Two panels. On the left, a shaded square cell with a dot at its centre marked $\mathbf{y}$; a thick red arrow labelled force rises from the centre, and four blue arrows labelled moment point outward from the cell's faces. A long arrow labelled $\mathbf{r}$ equals $\mathbf{x}$ minus $\mathbf{y}$ runs from the cell to a red dot marked $\mathbf{x}$ at the upper right, from which a red arrow labelled displacement leaves and above which a small dashed square deformed into a parallelogram is labelled strain. A boxed line below reads: the integral of G times the y -derivative of m equals minus the integral of the y -derivative of G times m , equals plus the integral of the x -derivative of G times m . On the right, a two-by-two grid of shaded blocks with rows labelled u , three rows, and strain, six rows, and columns labelled f , three columns, and m , six columns; the blocks read G , the first derivative of G , the first derivative of G , and the second derivative of G ; an arrow down the right side reads rows: one receiver derivative down, and an arrow along the bottom reads columns: one receiver derivative across.

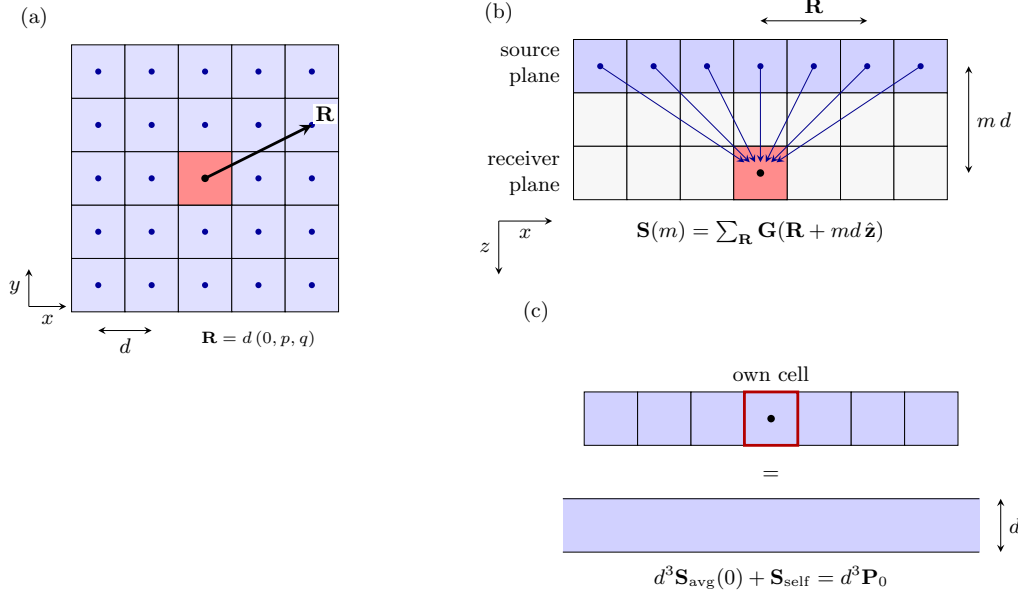


Figure 2: The lattice and its sums. (a) Plan view of one plane of voxels: their centres form a square lattice of pitch d , and a lattice vector $\mathbf{R} = d(0, p, q)$ joins the centre of one voxel (shaded darker) to another. (b) Cross-section (z down), between planes: the voxel of a receiver plane (shaded darker) is coupled to every voxel of a source plane m planes above it, one for each lattice vector $\mathbf{R}$; the sum over $\mathbf{R}$ is the specular lattice sum of Eq. (6). (c) Within a plane: the sum over every cell of the plane, the receiver’s own cell (outlined) included, is the field of a uniform plate of thickness d , the tiling identity of Eq. (13).

Alt text: Three schematic panels. On the left, a plan view of a five-by-five grid of square voxels with a dot at each centre; the centre voxel is highlighted, an arrow labelled $\mathbf{R}$ runs from its centre to another voxel centre, and the spacing between neighbouring centres is marked d . At the top right, a cross-section through three rows of voxels: arrows from the centre of every voxel in the top row converge on a highlighted voxel two rows below, with the plane separation $m d$ marked. Below it, a single row of voxels with one outlined as the receiver’s own cell is shown equal to a continuous shaded band of the same thickness d , a uniform plate.

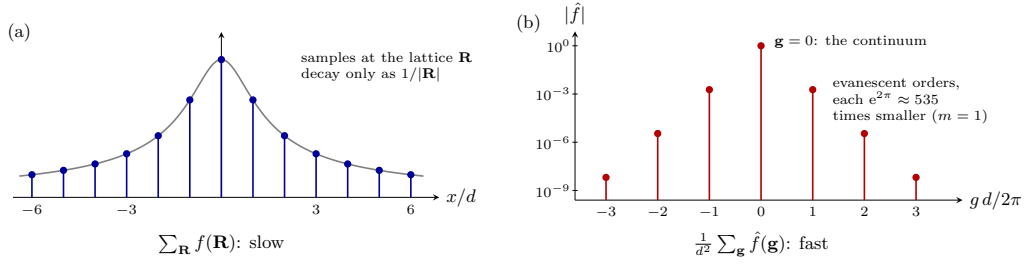


Figure 3: The Poisson summation formula (9) applied to a plane of voxels, along one lateral axis. (a) Real space: the field of each voxel of a source plane at a receiver one plane away, sampled at the lattice points; the samples decay only as $1/|\mathbf{R}|$, so their sum converges slowly. (b) The same sum after Poisson summation, on a logarithmic scale: samples of the lateral Fourier transform at the reciprocal-lattice orders $\mathbf{g} = 2\pi p/d$. The order $\mathbf{g} = 0$ is the continuum's plane wave; every other order is evanescent and, at $m = 1$, smaller than its neighbour by $e^{2\pi} \approx 535$. Magnitudes are schematic: the static decay $e^{-|\mathbf{g}|md}$ is shown.

Alt text: Two panels. On the left, a smooth bell-shaped curve that decays slowly on both sides is sampled by vertical stems at integer positions from minus six to six, the stems remaining clearly visible far from the centre. On the right, on a logarithmic vertical axis from one down to ten to the minus nine, a stem of height one at zero is labelled as the continuum, and stems at plus and minus one, two and three fall by about three powers of ten per step and are labelled as evanescent orders.

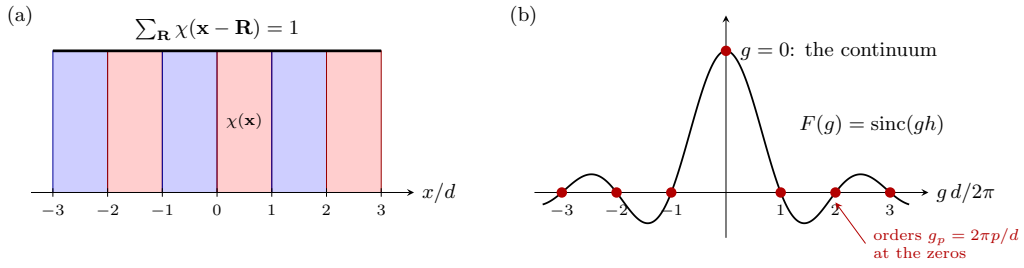


Figure 4: Why cubes make the coupling exact. (a) Cubes tile the plane: the indicator functions χ of the translated cells sum to exactly one everywhere, a partition of unity. (b) The same fact in Fourier space: the cube's lateral form factor $F(g) = \text{sinc}(gh)$ (shown along one axis) vanishes at every reciprocal-lattice order $g_p = 2\pi p/d$ with $p \neq 0$, so of all the orders in the lattice sum (10) only $g = 0$, the continuum's plane wave, survives the source-cell average.

Alt text: Two panels. On the left, adjacent box functions of unit height and unit width, alternately shaded, lie side by side along an axis marked in voxel widths, and a thick horizontal line at height one marks their sum. On the right, the sinc curve, equal to one at the origin and oscillating with decaying amplitude, is plotted against wavenumber in units of $2\pi/d$; dots mark the reciprocal-lattice orders, one at height one at the origin and all the others on the axis at the zeros of the curve.

868 9. A heterogeneous layer

869 A heterogeneous medium is not a layer of identical cells. The simplest
870 heterogeneous case with an exact answer is a stratified model built from
871 layers of constant properties, with real interfaces between them: the form of
872 many layered crustal velocity models. The last test takes such a model. A
873 layer $D = 4$ m thick is divided into eight model layers of 0.5 m, and in each
874 one $\Delta\lambda$, $\Delta\mu$ and $\Delta\rho$ are drawn uniformly within $\pm 15\%$ of the background
875 (the values are listed in `ContinuumLimit_Heterogeneous.w1`). The result is
876 a random stratification with physical contrasts. The voxels refine the model's
877 layering, with m planes of voxels in each model layer, so no voxel straddles
878 a model interface: the natural discretisation when the model's interfaces are
879 known. At $\omega = 1500 \text{ rad s}^{-1}$ the layer is $k_P D = 1.2$ and $k_S D = 2.0$ thick, so
880 the field reverberates between the model interfaces. The schemes are those
881 of §§7–8, with the contrast now changing from plane to plane; nothing else
882 in them changes. In the terms of the experiment of §7, the test is this.

- 883 1. *The medium.* The eight sets of contrasts are drawn once, from a fixed
884 seed, and the same eight are used in every run, so that every grid and
885 every scheme sees the same model.
- 886 2. *Normal incidence.* The source is the plane force at $z_s = -12$ m, and the
887 scattered displacement is taken at $z = -1$ m, where it is the reflected
888 wave, and at $z = D + 2 = 6$ m, where it is the change in the transmitted
889 wave. The P problem has the modulus contrast $\Delta\lambda + 2\Delta\mu$ and the wave
890 speed α ; the S problem is the same one-dimensional problem with $\Delta\mu$
891 and β .
- 892 3. *Oblique incidence.* A P plane wave is incident at 20° to the vertical.

893 What is measured is the pair of displacement amplitude vectors of
894 the reflected P and the reflected S plane waves, R_{PP} and R_{PS} , in the
895 specular order. The lateral sums of the coupling (§8) run over the
896 reciprocal-lattice orders $|p|, |q| \leq 8$.

897 4. *The exact answer.* The state of displacement and traction, two compo-
898 nents at normal incidence and four for P–SV at 20° , is carried across
899 each model layer by the matrix exponential of that layer’s system ma-
900 trix times its thickness, the eight matrices are multiplied in order,
901 and the radiation conditions of the two half-spaces are imposed on
902 the product. The arithmetic carries 30 to 40 digits. The reference
903 shares nothing with the voxel scheme but the model. It is checked
904 against a third algorithm, the reflection–transmission recursion of Ken-
905 nett [49] over the nine interfaces: the same-type coefficients, R_{PP} and
906 R_{SS} at normal incidence and R_{PP} at 20° , agree to 10^{-14} (`scripts/`
907 `gate_heterogeneous_reference_vs_kennett.py`). Only same-type
908 coefficients are compared, because they do not depend on how the up-
909 and downgoing waves are normalised; a converted coefficient does. One
910 sign differs, by convention: at normal incidence the one-dimensional
911 model’s R_{PP} is a ratio of vertical displacements, the recursion’s a ra-
912 tio of amplitudes along each wave’s own direction, and the reflected P
913 wave travels upwards.

914 5. *The error.* As in §7, the difference from the exact scattered quan-
915 tity relative to its size: for the amplitude vectors at 20° , $|\mathbf{R}_{\text{scheme}} -$
916 $\mathbf{R}_{\text{exact}}|/|\mathbf{R}_{\text{exact}}|$.

917 6. *The refinement.* Each model layer is cut into m planes of voxels, so

918 that the layer holds $n = 8m$: $m = 1, 2, 4, 8, 16$ at normal incidence
919 and $m = 1, 2, 4$ at 20° , where the lateral sums make each run dearer.
920 The orders are apparent orders, from $m = 4$ to 16 and from $m = 2$ to
921 4.

The errors are in Table 11 and Fig. 6(c).

Table 11: Relative error against the exact stratified layer, for m planes of voxels per model layer (§9). At normal incidence the order is measured from $m = 4$ to 16, where it is also 2.00 for collocation and 4.00 in transmission; at 20° it is measured from $m = 2$ to 4.

		$m = 1$	$m = 2$	$m = 4$	order
P, normal, R	mean only	1.9×10^{-3}	4.7×10^{-4}	1.2×10^{-4}	2.00
	first moment	8.5×10^{-7}	5.3×10^{-8}	3.3×10^{-9}	4.00
S, normal, R	mean only	5.3×10^{-3}	1.3×10^{-3}	3.3×10^{-4}	2.00
	first moment	6.2×10^{-6}	3.9×10^{-7}	2.4×10^{-8}	4.00
P, 20° , R_{PP}	mean only	1.4×10^{-3}	3.6×10^{-4}	9.0×10^{-5}	2.00
	first moments	3.3×10^{-7}	2.1×10^{-8}	1.3×10^{-9}	4.00
P, 20° , R_{PS}	mean only	2.8×10^{-3}	6.9×10^{-4}	1.7×10^{-4}	2.00
	first moments	1.4×10^{-6}	8.8×10^{-8}	5.5×10^{-9}	4.00

922
923 *The interfaces cost nothing.* Every order is the one measured for the uniform
924 layer. At a model interface the strain of the true field jumps. A voxel basis
925 that is discontinuous from voxel to voxel represents that jump exactly when
926 the interface lies on a voxel boundary, so the interface adds no error of its
927 own. The only requirement is that voxel boundaries coincide with the model's
928 interfaces. A voxel that straddled an interface would carry a contrast that is
929 not uniform within it, a case not examined here.

930 *What fourth order buys.* For a reflection error of 10^{-6} at normal incidence
931 (P), the mean-only voxel needs 347 planes of voxels, about 43 per model
932 layer. That figure is not a run: it is the second-order law carried from the
933 finest grid computed, 128 planes, whose error is 7.3×10^{-6} , to the target,
934 $n = 128 (7.3 \times 10^{-6} / 10^{-6})^{1/2}$. The first-moment voxel needs 8, one per model
935 layer, and that figure is a run: the model's own layering already gives $8.5 \times$
936 10^{-7} . Counting unknowns (9 per voxel against 36), that is 3123 against 288
937 per vertical column of voxels, a factor of eleven. In three dimensions, with
938 the lateral grid refined as the vertical one is, the number of voxels falls by
939 $(347/8)^3 \approx 8 \times 10^4$ and the number of unknowns by about 2×10^4 . The same
940 model has been solved independently in Python, with a contrast for each
941 plane: the raw reflected displacements agree with the Mathematica scheme
942 to 1.1×10^{-15} at normal incidence and at 20° , with and without the first
943 moments. Lateral heterogeneity needs the full three-dimensional solver; it is
944 the first extension discussed below.

945 *A medium that varies within a cell.* The stratified model is exact on its own
946 grid: every cell is homogeneous. A smoothly varying medium is not, and
947 sampling it as cells of constant properties is an approximation of its own.
948 Let the contrast of cell j be its Legendre projection to degree r , $\Delta(z) =$
949 $\Delta_0 \sum_{c \leq r} f_c^{(j)} P_c(s/h)$, with the voxel's field of degree p as in §8. The product
950 of contrast and field is again a Legendre sum, $P_c P_b = \sum_e A_{cbe} P_e$, with $A_{cbe} =$
951 $(2e+1) \begin{pmatrix} c & b & e \\ 0 & 0 & 0 \end{pmatrix}^2$ the square of a Wigner $3j$ symbol, so the coupling of Eq. (39)

952 becomes

$$\iint_j \phi_a(z) \mathbf{K}(z-z') \Delta(z') \phi_b(z') dz dz' = \sum_{c \leq r} \sum_{e \leq p+r} f_c^{(j)} A_{cbe} \left[\iint_j \phi_a(z) \mathbf{K}(z-z') \phi_e(z') dz dz' \right] \Delta_0 : \quad (43)$$

953 the same kernel moments, now to degree $p+r$, and nothing else new. Against
 954 the exact graded layer, a high-precision solution of $(M(z)u')' + \omega^2 \rho(z)u = 0$,
 955 the order of convergence is

$$\min(2p+2, 2r+2), \quad (44)$$

956 or $2p+2$ when degree r represents the profile exactly (Table 12; `ContinuumLimit_`
 957 `GradedContrast.wl`). A contrast that is constant in each cell therefore
 958 caps every voxel at second order, however rich its field basis, and for the
 959 smooth profile the voxel of degree 2 with a linear contrast has the error
 960 of the voxel of degree 1 to within 0.06%. At first order in the contrast
 961 this is what one expects: the contrast's projection error is orthogonal to
 962 the polynomials of degree r , so against the smooth product of the inci-
 963 dent and adjoint fields it contributes at $O(h^{2r+2})$. To keep order $2p+2$
 964 the contrast must be represented to the degree of the field. The closed forms
 965 of the construction are tested one by one by an independent implementa-
 966 tion (`scripts/crosscheck_graded_contrast.py`): the Legendre moments
 967 of the kernel against $2h i^n j_n(ch)$ and 30-digit quadrature (to 2×10^{-16}), the
 968 linearisation coefficients against the $3j$ symbols (exactly), and the same-cell
 969 double integrals against 30-digit quadrature (1.4×10^{-16}). It then assem-
 970 bles the scheme without linearisation, evaluating the projected contrast at
 971 the quadrature nodes, against its own 30-digit solution of the graded layer;
 972 its errors agree with those of Table 12 to 10^{-3} , and at $\omega = 1200 \text{ rad s}^{-1}$ it

Table 12: A contrast that varies within the cells: the field of each voxel to Legendre degree p , its contrast to degree r . Relative reflection error at $n = 16$ planes and the order ($n = 4 \rightarrow 8 \rightarrow 16$), $\omega = 300 \text{ rad s}^{-1}$, for the profiles $1 + (2z/D - 1)/2$ (linear) and $1 + \frac{1}{2} \sin(2\pi z/D)$ (smooth) times the contrast of §7. Predicted: Eq. (44), and $2p + 2$ for the linear profile once $r \geq 1$.

p	r	linear profile		smooth profile	
		error	order	error	order
0	0	7.7×10^{-5}	2.00		
1	0	7.6×10^{-5}	2.00	5.5×10^{-5}	1.95
1	1	6.4×10^{-11}	4.00	1.4×10^{-7}	3.95
2	1	2.3×10^{-17}	6.00	1.4×10^{-7}	3.95
2	2			1.6×10^{-10}	5.98

973 measures the orders of Eq. (44) again.

974 *Which basis sets the error.* Equation (44) gives the order. The size of the
975 error, and which of the two bases it belongs to, appear when the scattered
976 field is separated by its order in the contrast: the Born term T_1 , and the
977 second-order term T_2 in which the field scattered once is scattered again.

978 *The scattering series.* Multiply the whole contrast, $\Delta\lambda$, $\Delta\mu$ and $\Delta\rho$
979 together, by a number s , and let $u(s)$ be the scattered displacement at an
980 observer at depth z_o outside the layer. It vanishes at $s = 0$ and is analytic
981 in s there, and its expansion is the Born series,

$$u(s) = sT_1 + s^2T_2 + s^3T_3 + \dots, \quad (45)$$

982 so that T_n is the part of the field at the full contrast, $s = 1$, that has been
983 scattered n times.

984 *The first two terms of the exact layer.* They are integrals over the layer.
 985 With the incident state $w^0 = (u^0, \varepsilon^0)$, the contrast $\Delta_1 = \text{diag}(\omega^2 \Delta \rho, \Delta M)$
 986 with profile $f(z)$, and the kernel $\mathbf{K}$ of Eq. (39), whose first row $(g, g')(z_o - z)$
 987 gives the displacement at the observer,

$$T_1 = \int_0^D (g, g')(z_o - z) \Delta_1 f(z) w^0(z) dz, \quad (46)$$

988

$$w^1(z) = \int_0^D \mathbf{K}(z - z') \Delta_1 f(z') w^0(z') dz', \quad T_2 = \int_0^D (g, g')(z_o - z) \Delta_1 f(z) w^1(z) dz. \quad (47)$$

989 Here w^1 is the state scattered once, inside the layer; its strain includes the
 990 local term of the kernel, $-f(z) \Delta M \varepsilon^0(z)/M_P$. Each further term repeats
 991 the step, $w^n(z) = \int_0^D \mathbf{K}(z - z') \Delta_1 f(z') w^{n-1}(z') dz'$. For a plane force at z_s
 992 above the layer, $w^0(z) = c e^{ik(z-z_s)}(1, ik)$ with $c = i/(2M_P k)$, every integrand
 993 is an exponential times the profile. Writing $a = \omega^2 \Delta \rho$ and $b = \Delta M$, and
 994 carrying out the products of Eqs. (46) and (47), the terms seen in reflection
 995 ($z_o < 0$) and in transmission ($z_o > D$) are, for any profile,

996

$$\begin{aligned} T_1^R &= c^2 e^{-ik(z_o + z_s)} (a + bk^2) \int_0^D f(z) e^{2ikz} dz, & T_1^T &= c^2 e^{ik(z_o - z_s)} (a - bk^2) \int_0^D f(z) dz, \\ T_2^R &= c^2 e^{-ik(z_o + z_s)} \left[c (a^2 - b^2 k^4) \int_0^D \int_0^D f(z) f(z') e^{ik(z+z'+|z-z'|)} dz' dz - \frac{b^2 k^2}{M_P} \int_0^D f(z)^2 e^{2ikz} dz \right], \\ T_2^T &= c^2 e^{ik(z_o - z_s)} \left[c (a - bk^2)^2 \int_0^D \int_0^z f(z) f(z') dz' dz + c (a + bk^2)^2 \int_0^D \int_z^D f(z) f(z') e^{2ik(z'-z)} dz' dz \right. \\ &\quad \left. + \frac{b^2 k^2}{M_P} \int_0^D f(z)^2 dz \right]. \end{aligned} \quad (49)$$

997 The first-order reflection is the Fourier transform of the profile at the wavenum-
 998 ber $2k$, and the first-order transmission its integral. In each second-order

term the last integral is the local term of the kernel, in which both scatterings are at one point and the profile appears squared; the others are the two scatterings at different depths. In transmission the wave scattered twice forwards carries the contrast of velocity twice, $(a - bk^2)^2$, and the wave scattered backwards and then forwards again the contrast of impedance twice, $(a + bk^2)^2$. For the uniform layer ($f = 1$) the integrals give

$$T_1^R = c^2 e^{-ik(z_o+z_s)} (a+bk^2) \frac{e^{2ikD} - 1}{2ik}, \quad T_1^T = c^2 e^{ik(z_o-z_s)} (a-bk^2) D, \quad (50)$$

$$\begin{aligned} T_2^R &= \frac{c^2 e^{-ik(z_o+z_s)}}{4k^3 M_P} \left[-i(a^2 + b^2 k^4) + e^{2ikD} (a^2(i + 2kD) + b^2 k^4(i - 2kD)) \right], \\ T_2^T &= \frac{i c^2 e^{ik(z_o-z_s)}}{8k^3 M_P} \left[a^2 (1 - e^{2ikD} + 2kD(i + kD)) - 2abk^2 (e^{2ikD} - 1 + 2kD(kD - i)) \right. \\ &\quad \left. + b^2 k^4 (1 - e^{2ikD} + 2kD(kD - 3i)) \right]. \end{aligned} \quad (51)$$

The reflection sees the contrast of impedance, $a + bk^2 = k^2 M_P (\Delta\rho/\rho + \Delta M/M_P)$, and the transmission that of velocity, $a - bk^2$. As $kD \rightarrow 0$, $T_2^R \rightarrow -c^2 e^{-ik(z_o+z_s)} b^2 k^2 D/M_P$: the local term alone, which is the starting point of the argument below. For the smooth profile the integrals of Eqs. (48) and (49) are evaluated in closed form symbolically, and by quadrature they return the same values to 10^{-16} . Equations (50) and (51) equal, identically, the first two Taylor coefficients in s of the exact reflection and transmission of the layer of §3 (`ContinuumLimit_BornTerms.wl`). At $\omega = 300 \text{ rad s}^{-1}$, $|T_2|/|T_1|$ is 0.039 in reflection and 0.17 in transmission for the uniform layer.

The first two terms of the scheme. Write Eq. (39) as $(\mathbf{M} - \mathbf{K}) \mathbf{c} = \mathbf{b}$, with $\mathbf{M}$ the diagonal matrix of the $\int_i \phi_a^2 dz$, $\mathbf{K}$ the matrix of the coupling blocks times Δ_1 , and $\mathbf{b}$ the moments of the incident field, and let $\mathbf{r}$ be the row that

1018 gives the scattered displacement at the observer from the coefficients, $u = \mathbf{r} \mathbf{c}$.
 1019 Scaling the contrast by s scales $\mathbf{K}$ and $\mathbf{r}$ by s , so $u(s) = s \mathbf{r} (\mathbf{M} - s \mathbf{K})^{-1} \mathbf{b}$ and

$$T_1^{\text{scheme}} = \mathbf{r} \mathbf{M}^{-1} \mathbf{b}, \quad T_2^{\text{scheme}} = \mathbf{r} \mathbf{M}^{-1} \mathbf{K} \mathbf{M}^{-1} \mathbf{b}. \quad (52)$$

1020 $\mathbf{M}^{-1} \mathbf{b}$ is the projection of the incident field onto each cell's functions: the first
 1021 term needs no solve. Splitting $\mathbf{K}$ into its blocks within a cell and its blocks
 1022 between cells splits T_2^{scheme} into the part scattered twice in one cell and the
 1023 part scattered in two; the same split of the inner integral of Eq. (47) splits
 1024 the exact T_2 .

1025 *A check by differences.* The terms can also be read from solutions at four
 1026 scaled contrasts, $s = \pm\epsilon$ and $\pm 2\epsilon$, with no knowledge of the formulas above.
 1027 The odd and even parts of u in s separate the odd orders from the even,
 1028 and the two step sizes remove the next term of each (the five-point central
 1029 differences for $u'(0)$ and $\frac{1}{2}u''(0)$):

$$\begin{aligned} T_1 &= \frac{8[u(\epsilon) - u(-\epsilon)] - [u(2\epsilon) - u(-2\epsilon)]}{12\epsilon} + 4\epsilon^4 T_5 + \dots, \\ T_2 &= \frac{16[u(\epsilon) + u(-\epsilon)] - [u(2\epsilon) + u(-2\epsilon)]}{24\epsilon^2} + 4\epsilon^4 T_6 + \dots \end{aligned} \quad (53)$$

1030 A negative s is a medium slower and lighter than the background by the
 1031 same fractions, as legitimate a medium as the other. With $\epsilon = 0.01$ the
 1032 remainders $4\epsilon^4 T_5/T_1$ and $4\epsilon^4 T_6/T_2$ are below 2×10^{-12} for this layer, and
 1033 the formulas applied to the exact layer return Eqs. (50) and (51) less exactly
 1034 those remainders, to 10^{-17} . The measurements below use the differences for
 1035 the scheme and the exact layer alike, each in its own arithmetic, and they
 1036 agree with Eqs. (46), (47) and (52).

1037 *Why the terms must be separated.* The law below is a statement about T_2 .
 1038 At the full contrast the terms of third and higher order are 7% of the second

1039 and carry errors of their own, and if every order above the first is taken
 1040 together the ratio of Eq. (55) reads 0.90 to 0.94 where the second-order term
 1041 alone gives 1.00 to 1.01.

1042 With $f(z)$ the profile and Π_q the L^2 projection onto the Legendre poly-
 1043 nomials of degree at most q in each cell, define the *relative projection error*
 1044 of the medium at degree q ,

$$\mathcal{E}_q = \frac{\sum_{\text{cells}} \int (f - \Pi_q f)^2 dz}{\int f^2 dz} = 1 - \frac{\|\Pi_q f\|^2}{\|f\|^2}, \quad \|f\|^2 = \int f^2 dz : \quad (54)$$

1045 the fraction of the squared norm of the profile that polynomials of degree q
 1046 on the cells cannot represent. It is a property of the medium and the grid
 1047 alone; it involves neither the wave nor the solution. For the smooth profile
 1048 and every one of the nine pairs with $p, r \leq 2$, the relative error of T_2 is

$$\frac{|T_2^{\text{scheme}} - T_2|}{|T_2|} = \mathcal{E}_{\min(p,r)} \quad (55)$$

1049 to within 1.3% at $\omega = 300 \text{ rad s}^{-1}$ on 4 to 16 cells: the ratio of the two sides
 1050 is 0.998 to 1.005 when $\min(p, r)$ is 0 or 1, and 1.010 to 1.013 when it is 2.
 1051 The error of the Born term is smaller by two to six orders of magnitude. The
 1052 error of the whole is therefore that of T_2 , the projection error times $|T_2|/|T_1|$
 1053 (0.044 here): for $p = r = 1$ on 16 cells that predicts 1.6×10^{-7} , and Table 12
 1054 has 1.4×10^{-7} .

1055 Equation (55) is exact in the long-wave limit, for a reason that shows what
 1056 each basis is for. The strain-strain entry of the kernel, $g'' = -k^2 g - \delta(z)/M_P$,
 1057 is local as $kD \rightarrow 0$, and the incident strain ε_0 is then uniform across the layer,
 1058 so the strain scattered once is $-(\Delta M/M_P) f(z) \varepsilon_0$: it is the profile itself. A
 1059 cell holds the contrast as $\Pi_r f$ and the field to degree p , so it holds the once-
 1060 scattered strain as $\Pi_p(\Pi_r f) = \Pi_q f$ with $q = \min(p, r)$. The second-order

1061 term pairs the contrast with that strain:

$$T_2 \propto \int f \cdot f \, dz = \|f\|^2, \quad T_2^{\text{scheme}} \propto \int (\Pi_r f)(\Pi_q f) \, dz = \|\Pi_q f\|^2, \quad (56)$$

1062 the last because $\Pi_q f$ lies in the range of Π_r . The relative error is $1 -$
 1063 $\|\Pi_q f\|^2 / \|f\|^2 = \mathcal{E}_q$. Because the kernel is local in this limit, the second
 1064 scattering takes place in the cell of the first: T_2 is the second-order term of
 1065 the single-site T -matrix, and Eq. (55) is a statement about the single site.
 1066 The T -matrix sums the rescattering within a cell exactly for the cell as it
 1067 is represented, a contrast $\Pi_r f$ and a field of degree p ; its error is that the
 1068 cell is not the medium. The split is measured directly (`scripts/measure_`
 1069 `layer_bases.py`): at $\omega = 300 \text{ rad s}^{-1}$ on 4, 8 and 16 cells, T_2 of the scheme
 1070 and of the exact layer are each divided into the part whose two scatterings
 1071 lie in one cell and the part between different cells. For all nine pairs the
 1072 error of the same-cell part is 0.998 to 1.013 times $|T_2| \mathcal{E}_{\min(p,r)}$, and that of
 1073 the part between cells at most 0.002 times it. The coupling between cells
 1074 contributes only the departure from the limit. Measured, the ratio of the
 1075 two sides of Eq. (55) for $p = r = 1$ on 8 cells departs from one by 1.7×10^{-2} ,
 1076 4.3×10^{-3} , 4.8×10^{-4} and 4×10^{-5} at $kD = 0.24, 0.12, 0.04$ and 0.012 : as
 1077 $(kD)^2$ (`scripts/measure_layer_bases.py`). The pairing in Eq. (56) is ver-
 1078 ified symbolically for all nine pairs, and the projection errors are computed
 1079 exactly (`ContinuumLimit_LayerDefect.wl`); they fall by 4^{q+1} for each halv-
 1080 ing of the cells.

1081 Three consequences follow. The two bases must be matched: a contrast
 1082 of higher degree than the field is wasted (the pairs $(p, r) = (0, 1)$ and $(0, 2)$
 1083 have the errors of $(0, 0)$ to three digits), and a field of higher degree than
 1084 the contrast improves only the Born term. The wavefield basis is there to

1085 carry what the medium imprints on the field, not only the wave: at these
 1086 frequencies the wave itself varies little across a cell. And the error is known
 1087 from the medium, before anything is solved: $\mathcal{E}_q$ is a property of the profile
 1088 and the cells. It vanishes when the medium is piecewise constant on the cells,
 1089 which is the stratified model above; there the contrast needs no more than a
 1090 constant in each cell, and the degree of the field alone sets the order.

1091 *A stratified background.* Everything above takes the background to be a
 1092 whole space. The method is meant for a background that is plane strati-
 1093 fied, with the voxels carrying the departure from it, and the results must
 1094 hold there. They do, in the one case tested: normal incidence, P, with the
 1095 background two half-spaces joined at $z = z_I$ below the layer, the lower one
 1096 faster and denser (displacement reflection coefficient -0.19). The voxels then
 1097 interact through the Green's function of that background,

$$g(z, z') = \frac{i}{2M_P k} \left[e^{ik|z-z'|} + \rho_I e^{ik(2z_I - z - z')} \right], \quad z, z' < z_I, \quad (57)$$

1098 with ρ_I the reflection coefficient of the interface: the whole-space term and
 1099 a reverberation that is smooth and carries no local term. The incident field
 1100 is the field of the source in the background, and the exact answer is the
 1101 reflection of the whole stack less that of the background. With the interface
 1102 3m below the layer, and again with the lowest voxel against it, the scheme
 1103 converges at order 2.00, 4.00 and 6.0 for $p = 0, 1$ and 2 on the uniform
 1104 layer, and at 1.98, 3.98 and 5.98 on the smooth profile with $r = p$. The
 1105 first-order term is predicted to round-off (5×10^{-16}): it is the sum over
 1106 four paths, the incident wave travelling down or up and the scattered wave
 1107 leaving up or down, of the exact first-order term of each path times the

1108 factor $\sum_{a \leq p} (2a + 1) j_a(k_{\text{in}} h) j_a(k_{\text{out}} h) / j_0((k_{\text{in}} - k_{\text{out}}) h)$ for that path's pair
 1109 of wavenumbers, whose leading terms are the constants of Eq. (41). And the
 1110 relative error of T_2 over $\mathcal{E}_p$ is 1.00 to 1.02 at $kD = 0.12$ on 4 to 16 cells for all
 1111 three degrees, departing from one by 1.1×10^{-4} and 3×10^{-5} at $kD = 0.012$
 1112 for the two positions of the interface: Eq. (55) holds as in the whole space
 1113 (`scripts/measure_layer_stratified_background.py`; the exact stack is
 1114 computed in 30-digit arithmetic). The reason is the one given for Eq. (56):
 1115 the reverberation adds no local term, so the second scattering in the cell of
 1116 the first is unchanged. This check is by one implementation against an exact
 1117 solution.

1118 *Oblique incidence, and a three-layer background.* At oblique incidence the
 1119 test is the three-dimensional scheme of Eq. (42), with P, SV and SH waves, in
 1120 two backgrounds: the two half-spaces above, and a three-layer background in
 1121 which the voxels lie in a layer between a slower half-space above ($\alpha = 4200$,
 1122 $\beta = 2400 \text{ m s}^{-1}$, $\rho = 2300 \text{ kg m}^{-3}$) and the faster one below (6500, 3700,
 1123 2800), so that the layer reverberates between two interfaces. The experiment
 1124 is built in four steps.

1125 *The kernel.* At each lateral wavenumber κ the whole-space kernel between
 1126 two depths factorises over the plane-wave modes, three going down and three
 1127 going up (P, SV, SH): $\hat{\mathbf{G}}(\kappa; z - z') = \mathbf{D}_{\downarrow} \mathbf{E}(z - z') \mathbf{S}_{\downarrow}$ for $z > z'$, with $\mathbf{S}_{\downarrow}$ the
 1128 3×9 matrix of mode amplitudes radiated downwards by a nine-component
 1129 source, $\mathbf{E}(L) = \text{diag}(e^{i\gamma_m L})$ the propagation over a distance L with γ_m the
 1130 vertical wavenumber of mode m , and $\mathbf{D}_{\downarrow}$ the 9×3 matrix of the modes'
 1131 states; and likewise upwards. With the interfaces at z_A above and z_B below,
 1132 $H = z_B - z_A$, and $\mathbf{R}_A, \mathbf{R}_B$ the 3×3 reflection matrices of the two interfaces

1133 seen from inside the layer, a source at z' sends $\mathbf{s}_\downarrow = \mathbf{S}_\downarrow q$ down and $\mathbf{s}_\uparrow = \mathbf{S}_\uparrow q$
 1134 up, and the reverberation it sets up is

$$\begin{aligned} \mathbf{d}_A &= (\mathbf{I} - \mathbf{R}_A \mathbf{E}(H) \mathbf{R}_B \mathbf{E}(H))^{-1} \mathbf{R}_A [\mathbf{E}(z' - z_A) \mathbf{s}_\uparrow + \mathbf{E}(H) \mathbf{R}_B \mathbf{E}(z_B - z') \mathbf{s}_\downarrow], \\ \mathbf{u}_B &= \mathbf{R}_B [\mathbf{E}(z_B - z') \mathbf{s}_\downarrow + \mathbf{E}(H) \mathbf{d}_A], \\ \hat{\mathbf{G}}_{\text{rev}}(\boldsymbol{\kappa}; z, z') q &= \mathbf{D}_\downarrow \mathbf{E}(z - z_A) \mathbf{d}_A + \mathbf{D}_\uparrow \mathbf{E}(z_B - z) \mathbf{u}_B, \end{aligned} \tag{58}$$

1135 with $\mathbf{d}_A$ the downgoing amplitudes leaving the upper interface and $\mathbf{u}_B$ the
 1136 upgoing ones leaving the lower. It is added to the whole-space kernel in
 1137 every block of Eq. (42), at every order of the reciprocal lattice. It is smooth
 1138 and separable in z and z' , so its integrals over the cells are one-dimensional
 1139 moments of exponentials. For two half-spaces $\mathbf{R}_A = 0$ and only the second
 1140 line survives.

1141 *The interfaces.* $\mathbf{R}_A, \mathbf{R}_B$ and the transmission matrices into and out of the
 1142 layer follow from continuity of displacement and traction at each interface:
 1143 six equations in the states of the six modes on either side, solved for each
 1144 lateral wavenumber, evanescent orders included.

1145 *The incident field and the output.* A plane wave of unit amplitude arrives
 1146 from the upper half-space. Its field in the layer, a downgoing part and an
 1147 upgoing part with the reverberation summed as in Eq. (58), is the right-
 1148 hand side of the scheme. The output is the upgoing wave of the same type
 1149 as the incident one, arriving at z_A from the voxels and transmitted into the
 1150 upper half-space: the change that the layer of voxels makes to the same-type
 1151 reflection coefficient (R_{PP} , R_{SS} or R_{SH}), a quantity that does not depend on
 1152 how the waves are normalised.

1153 *The reference, and a check of the construction.* The exact change is the

1154 same-type reflection coefficient of the stack with the layer, less that of the
 1155 background, both from Kennett's recursion [49], an independent algorithm.
 1156 Before any voxel is introduced, the background's own reflection coefficient
 1157 computed from the modes and the interface matrices above agrees with Ken-
 1158 nett's to 5×10^{-15} in all twelve cases.

1159 The results are in Table 13, at $\omega = 3000 \text{ rad s}^{-1}$ ($k_S D = 2$; at 300 rad s^{-1}
 1160 the sixth-order errors lie below round-off, as in Table 10) and 20° in the
 1161 upper half-space. In all thirty-six cases the order is that of the whole space,
 1162 with the interfaces 3m from the layer and with the voxels against them
 1163 (`scripts/measure_oblique_stratified_background.py`). The errors are
 1164 close to those of the whole space at the same frequency (Table 10): the
 1165 background changes the field the voxels must represent, not how well they
 1166 represent it.

1167 *The two terms of the error at oblique incidence.* The orders say that the
 1168 scheme converges; the two-term analysis says what its error is. Both terms
 1169 are tested at 20° , in the whole space and in the stratified backgrounds, for
 1170 P, SV and SH waves.

1171 *The first-order term.* The scheme is written as $(\mathbf{M} - \mathbf{K})\mathbf{c} = \mathbf{b}$ with the
 1172 output $u = \mathbf{r}\mathbf{c}$, as in Eq. (52), so that $T_1^{\text{scheme}} = \mathbf{r}\mathbf{M}^{-1}\mathbf{b}$ with no differencing.
 1173 The prediction is built from plane waves. The background field in the layer
 1174 is a sum of six, P, SV and SH going down and going up, and the voxels
 1175 radiate into six. For a pair made of an incident wave of wavevector $\mathbf{k}_{\text{in}}$ and
 1176 a radiated wave of wavevector $\mathbf{k}_{\text{out}}$, a cubic voxel of half-side h whose field is

Table 13: The layer of voxels in a stratified background at 20° , $\omega = 3000 \text{ rad s}^{-1}$: relative error of the change of the same-type reflection coefficient on $n = 4$ planes, and the order over the last halving ($n = 2 \rightarrow 4$), for field degrees $p = 0, 1$ and 2 .

background	wave	$p = 0$	$p = 1$	$p = 2$
two half-spaces, interface 3 m below	P	6.1×10^{-3} (2.03)	2.7×10^{-6} (4.02)	2.3×10^{-9} (6.04)
	SV	1.5×10^{-2} (2.11)	2.6×10^{-5} (4.13)	3.7×10^{-8} (6.09)
	SH	1.6×10^{-2} (2.11)	2.7×10^{-5} (4.13)	4.1×10^{-8} (6.09)
two half-spaces, interface against the voxels	P	5.1×10^{-3} (2.04)	3.8×10^{-6} (4.04)	2.1×10^{-9} (6.00)
	SV	1.6×10^{-2} (2.10)	2.5×10^{-5} (4.14)	3.3×10^{-8} (6.11)
	SH	1.7×10^{-2} (2.10)	2.4×10^{-5} (4.15)	3.4×10^{-8} (6.11)
three layers, interfaces 3 m away	P	6.3×10^{-3} (2.03)	8.7×10^{-7} (4.04)	2.4×10^{-9} (6.06)
	SV	1.0×10^{-2} (2.10)	5.8×10^{-6} (4.08)	5.2×10^{-8} (6.11)
	SH	1.3×10^{-2} (2.10)	3.0×10^{-6} (3.95)	9.3×10^{-8} (6.10)
three layers, interfaces against the voxels	P	4.8×10^{-3} (2.04)	2.4×10^{-6} (4.04)	4.1×10^{-9} (6.03)
	SV	8.5×10^{-3} (2.09)	7.1×10^{-6} (4.01)	5.0×10^{-8} (6.09)
	SH	1.4×10^{-2} (2.10)	5.8×10^{-6} (3.92)	8.5×10^{-8} (6.12)

1177 held to total degree p gives the exact first-order term of that pair times

$$1 + E_p(\mathbf{k}_{\text{in}}, \mathbf{k}_{\text{out}}) = \frac{\sum_{|a| \leq p} \prod_{i=1}^3 (2a_i + 1) j_{a_i}(k_{\text{in},i}h) j_{a_i}(k_{\text{out},i}h)}{\prod_{i=1}^3 j_0((k_{\text{in},i} - k_{\text{out},i})h)}, \quad (59)$$

1178 the sum being over the Legendre products $P_{a_1}P_{a_2}P_{a_3}$ of the voxel's basis and
 1179 j_n the spherical Bessel functions: the numerator is the projection of the inci-
 1180 dent wave onto the basis paired with that of the outgoing wave, the denom-
 1181 inator the exact pairing. Its leading departure from one is $-d^2 \mathbf{k}_{\text{in}} \cdot \mathbf{k}_{\text{out}}/12$
 1182 for $p = 0$, with $d = 2h$, and along one axis it is the factor of Eq. (40). The
 1183 predicted first-order term is the sum over the thirty-six pairs of the exact
 1184 term of each pair, an integral of two exponentials over the voxel, times its
 1185 factor, the radiated waves then being carried to the observer by the reverber-
 1186 ation of Eq. (58) exactly. On one and two planes of voxels of degree 0, 1 and
 1187 2, at $\omega = 3000 \text{ rad s}^{-1}$, the scheme's first-order term equals this prediction
 1188 to 2×10^{-15} in all twelve combinations of background and wave type. The
 1189 first-order error is therefore known in closed form at oblique incidence, in a
 1190 stratified background, for every wave type.

1191 *The second-order term.* For the law of Eq. (55) the contrast must vary
 1192 within the voxels. Each plane of voxels is given the smooth profile of §9
 1193 in depth, projected to Legendre degree $r = p$, and the coupling is that of
 1194 Eq. (43) with the reverberation added. The second-order term of the scheme
 1195 is $T_2^{\text{scheme}} = \mathbf{r} \mathbf{M}^{-1} \mathbf{K} \mathbf{M}^{-1} \mathbf{b}$. The exact T_2 needs the exact stack with a layer
 1196 whose properties vary continuously in depth, at oblique incidence, for which
 1197 a recursion over homogeneous layers is not exact. It is computed by two

1198 independent algorithms. The first is invariant imbedding, the impedance
1199 march of Haines et al. [50]: the six-component state of displacement and
1200 traction obeys $dy/dz = A(z)y$, and with the four 3×3 blocks A_{ij} of A
1201 the impedance Y , which gives the traction from the displacement, obeys the
1202 Riccati equation

$$\frac{dY}{dz} = A_{21} + A_{22}Y - YA_{11} - YA_{12}Y. \quad (60)$$

1203 It is started below the stack with the impedance of the downgoing waves
1204 of the lower half-space, which is the radiation condition, marched upwards,
1205 exactly through the homogeneous gaps and by fourth-order Runge–Kutta
1206 with the medium sampled at the stages through the graded layer, and at
1207 the top the reflection matrix follows from continuity of displacement and
1208 traction with the modes of the upper half-space. Y stays bounded, so no
1209 growing exponential is formed. The second is the propagator: the same
1210 state is integrated across the graded layer as a 6×6 matrix, with $A(z) =$
1211 $F(z) \text{diag}(i\gamma_m) F(z)^{-1}$ built at each depth from the plane-wave modes of the
1212 medium there, F the matrix of the six modes' displacements and tractions
1213 and γ_m their vertical wavenumbers, and the radiation conditions of the two
1214 half-spaces are imposed on the result. For the graded layer the two agree
1215 to 10^{-12} in all twelve cases with 400 steps of the march, whose error falls
1216 sixteenfold for each doubling of the steps; for a uniform layer the propagator
1217 equals Kennett's recursion to 3×10^{-14} . T_2 then follows from Eq. (53) at
1218 $\epsilon = 0.1$, where halving ϵ changes it by 2×10^{-8} . The ratio of the relative
1219 error of T_2 to the relative projection error $\mathcal{E}_p$ is in Table 14. On four planes
1220 the law holds to 3% for $p = 0$ and 1 and to 6% for $p = 2$ at $k_S D = 0.2$, and to
1221 0.6% for all three degrees at $k_S D = 0.067$, in every background and for every

Table 14: The relative error of the second-order term over the relative projection error of the medium, at 20° , on $n = 4$ planes with $r = p$; the law of Eq. (55) predicts one in the long-wave limit.

		$k_S D = 0.2$			$k_S D = 0.067$		
background	wave	$p = 0$	$p = 1$	$p = 2$	$p = 0$	$p = 1$	$p = 2$
whole space	P	1.000	0.998	1.011	1.000	1.000	1.000
	SV	0.998	0.992	1.032	1.000	0.999	1.004
	SH	0.999	0.992	1.031	1.000	0.999	1.003
two half-spaces, interface 3 m below	P	1.010	1.009	1.019	1.001	1.001	1.001
	SV	1.012	1.008	1.046	1.001	1.001	1.005
	SH	1.027	1.025	1.059	1.003	1.003	1.006
three layers, interfaces 3 m away	P	1.004	1.003	1.013	1.000	1.000	1.000
	SV	0.979	0.973	1.002	0.997	0.996	1.000
	SH	1.007	1.004	1.035	1.001	1.001	1.004
three layers, interfaces against the voxels	P	1.002	1.001	1.011	1.000	1.000	1.000
	SV	0.995	0.988	1.018	0.999	0.999	1.002
	SH	1.004	1.002	1.032	1.000	1.000	1.003

1222 wave type: the departure falls about ninefold as the frequency falls threefold,
1223 as $(kD)^2$. On two planes the ratios for $p = 0$ and 1 lie within 0.03 of those
1224 in the table; for $p = 2$ they are lower, 0.89 to 0.99 at $k_S D = 0.2$ and 0.988 to
1225 0.999 at $k_S D = 0.067$. The law is loosest for the highest degree at the highest
1226 frequency, because the projection error of degree 2 is so small (1.6×10^{-5} on
1227 four planes) that the departure from the long-wave limit is a larger share of
1228 it. Both terms of the error are thus known in advance at oblique incidence
1229 and in a stratified background, as they are at normal incidence in the whole
1230 space (`scripts/measure_oblique_two_term.py`).

1231 Two limits stand. The oblique construction is implemented once, and
1232 its checks are the exact solutions. And the voxels here fill their plane, so
1233 that the image of a voxel in an interface it touches is the image of a whole
1234 plane; a body of finite lateral extent against an interface, where that image
1235 is singular, has not been run.

1236 10. A sphere: what the layer predicts in three dimensions

1237 The layer and the stratified layer are laterally uniform. A sphere is not,
1238 and elastic scattering by a sphere, homogeneous [47, 48] or radially varying
1239 [51], has an exact solution: it is the usual benchmark for volume-integral and
1240 discrete-dipole methods. This section solves two spheres with the full three-
1241 dimensional voxel scheme: the sphere is voxelised on an n_{sub}^3 grid, the Foldy-
1242 Lax system is solved with the cell-averaged coupling by an FFT-accelerated
1243 iterative solver, and a P wave is incident. For the graded sphere the voxel is
1244 the mean-only voxel of §8 in three dimensions: each cell carries the mean of its
1245 nine-component state, tested with the same constant, and the L^2 projection

1246 of its contrast, which is its cell mean; every cell that overlaps the sphere is
 1247 kept, and the cells are coupled by cell-to-cell double integrals of the Green's
 1248 tensor, one block per orbit of the cube group [4]. For the sharp sphere
 1249 the voxel is the package's cube T -matrix, which differs from the collocation
 1250 closure only through its far-field corrections (§6.6) and has the same order,
 1251 each cell keeping the contrast of its centre. The far field is computed from
 1252 the solution for the incident plane wave itself, each voxel's displacement and
 1253 strain carrying the plane-wave phase at its centre; adding to that phase a
 1254 linear variation about the scatterer's centre, as a Taylor expansion of the
 1255 incident field would, counts the wave's variation twice and leaves an error of
 1256 order $(ka)^2$ that no refinement removes. In the terms of the experiment of
 1257 §7, the test is this.

- 1258 1. *The body.* A sphere of radius $a = 10$ m (Fig. 5) in the background of
 1259 §7, carrying the contrast of that section, at $\omega = 150$ and 300 rad s^{-1} ,
 1260 for which $k_S a = 0.5$ and 1 and $k_P a = 0.3$ and 0.6 .
- 1261 2. *The incident wave.* A P plane wave travelling along $+z$.
- 1262 3. *The grid.* The cube of side $2a$ that circumscribes the sphere is cut into
 1263 n_{sub}^3 cells of side $2a/n_{\text{sub}}$; which of them are kept is stated with each
 1264 voxel above.
- 1265 4. *The solve.* The iteration is carried to a relative residual of 10^{-11} , far
 1266 below every discretisation error reported, so that the errors are the
 1267 scheme's and not the solver's.
- 1268 5. *The exact answer.* The partial-wave series of the sphere, summed to
 1269 order 10 at $k_S a = 0.5$ and 11 at $k_S a = 1$, and evaluated as the exact
 1270 scattered displacement at the observers, not as its asymptotic form.

- 1271 6. *What is measured.* The scattered displacement vector at nine scattering
1272 angles, equally spaced from 11° to 169° in the z - x plane: forward and
1273 backward scattering and everything between, P and S together.
- 1274 7. *The error.* The largest component of the difference between the scheme's
1275 far field and the exact displacement over the nine angles, divided by the
1276 largest component of the exact one. It is a maximum, not an average,
1277 so that no angle is hidden.
- 1278 8. *The distance.* The solution of the scheme is the field inside the voxels,
1279 and the scattered field outside follows from it at any distance. For the
1280 far-field test it is read with the asymptotic formula, which keeps the
1281 term in $1/r$ alone: a choice of this test, not a property of the scheme.
1282 The reference is the exact field, so the two differ by the near-field terms,
1283 of relative size $1/(kr)$, however good the scheme. The graded sphere
1284 is therefore observed at 5×10^8 radii, where those terms are 10^{-8} of
1285 the peak: at 5×10^4 radii they are 1.4×10^{-4} , which is 4% of the
1286 discretisation error at 12 voxels across and most of it at 24, and would
1287 be read as a failure to converge. The sharp sphere, whose errors are
1288 a hundred times larger, is observed at 5×10^4 radii. The near field is
1289 tested separately, with a readout that drops nothing (Table 15).

1290 *A sphere without a staircase.* A voxelised sphere with a sharp surface changes
1291 its shape with every grid. A sphere whose contrast falls smoothly to zero at
1292 its surface does not: the voxels that cross the surface carry a vanishing con-
1293 trast. The body here has a homogeneous core $r < a/2$ carrying the contrast
1294 of §7, and across the shell $a/2 < r < a$ the contrast falls to zero as the
1295 smoothstep $s(x) = 10x^3 - 15x^4 + 6x^5$, $x = (a - r)/(a - a/2)$, which vanishes

1296 like $(a - r)^3$ at the surface. Its exact scattering follows, order by order, from
 1297 the radial equations of Takeuchi and Saito [51], integrated across the shell
 1298 from the core's regular solutions and matched to outgoing waves at $r = a$.
 1299 The equations are derived symbolically from the equations of motion, and
 1300 the solution is checked against a sphere of homogeneous concentric shells
 1301 in closed form, by unitarity and reciprocity, and by an independent imple-
 1302 mentation that agrees to 2×10^{-11} (`TakeuchiSaito.wl`, `ContinuumLimit_`
 1303 `GradedSphere.wl`, `scripts/crosscheck_graded_sphere.py`). Each voxel
 1304 carries the mean of the contrast over its cell. Table 16 shows the error falling
 1305 at second order from $n_{\text{sub}} = 6$ to 24 at both frequencies (`scripts/pilot_`
 1306 `graded_voxel_sphere.py`): the apparent order between successive grids is
 1307 2.02 to 2.06 at $k_S a = 0.5$ and 2.00 at $k_S a = 1$, and the least-squares orders
 1308 of Fig. 6(d) are 2.045 ± 0.004 and 1.994 ± 0.004 . With 24 voxels across the
 1309 error is 1.8×10^{-4} and 8.3×10^{-4} . At $n_{\text{sub}} = 4$ a voxel is as wide as the
 1310 graded shell and the order is not yet asymptotic. This is the order the layer
 1311 predicts for the mean-only voxel, and the first test of that prediction on a
 1312 body with genuine three-dimensional structure.

1313 What makes the sequence this clean is that the voxel carries a single error.
 1314 The collocation voxel of the sharp sphere applied to this body, each cell at
 1315 the profile's value at its centre and only the cells whose centres lie inside the
 1316 sphere kept, carries two: its own internal-field error and the error of sampling
 1317 the profile, both $O(h^2)$. Its error falls from 1.0×10^{-1} at $n_{\text{sub}} = 4$ to 4.7×10^{-3} ,
 1318 2.6×10^{-3} , 6.4×10^{-4} and 2.3×10^{-4} at 6, 8, 12 and 16, apparent orders 7.6,
 1319 2.0, 3.5 and 3.6 at $k_S a = 0.5$, and at $k_S a = 1$ the apparent orders are 1.7, 2.5
 1320 and 2.4 from 6 to 16: on these grids its error is below the mean-only voxel's,

1321 but its apparent order is not that of any single term and cannot be read from
1322 them. The same collocation voxel with the cell mean of the contrast, and
1323 every overlapping cell kept, gives 3.4×10^{-3} , 1.9×10^{-3} and 8.2×10^{-4} at 6,
1324 8 and 12, apparent orders 2.03 and 2.05: the sampling of the profile was the
1325 second term, and with it removed the collocation and mean-only voxels differ
1326 only by their constants, as in Table 7. The cell mean, with every overlapping
1327 cell kept, represents the same body on every grid, and the order the layer
1328 predicts is then the one observed.

1329 The first-moment voxel of §8, extended to three dimensions with a con-
1330 trast linear in each cell, converges at fourth order on this same sphere [4]: its
1331 apparent orders are 4.01 and 4.13 where the discretisation error is isolated
1332 at weak contrast, and at the full contrast its error with 16 voxels across is
1333 4.9×10^{-6} and 5.9×10^{-6} , 82 and 317 times below the mean-only voxel's on
1334 the same grid.

1335 *The near field.* A receiver is seldom at infinity, and the scheme does not
1336 require it to be. The scattered field at a point $\mathbf{x}$ outside the voxels is the
1337 propagator applied to the source that each voxel carries, the contrast times
1338 the field found in it:

$$\psi^{\text{sc}}(\mathbf{x}) = \sum_j \int_{V_j} \mathbf{K}(\mathbf{x} - \mathbf{x}') \Delta(\mathbf{x}') e_j(\mathbf{x}') d\mathbf{x}', \quad (61)$$

1339 with $\mathbf{K}$ the nine-component Green's tensor of §2, the same one that couples
1340 the voxels in the solve, Δ the contrast operator of Eq. (5) and e_j the poly-
1341 nomial internal state of voxel j . Its first three rows are the displacement
1342 and its last six the strain. Nothing is dropped: the terms in $1/r^2$ and $1/r^3$
1343 are in $\mathbf{K}$. The integrand's source is a polynomial in each voxel and is in-

1344 tegrated by a tensor Gauss rule with six nodes a side; the rule's own error
 1345 is measured by repeating the readout with four nodes, and from 8 voxels
 1346 across the two differ by less than 4×10^{-6} at $1.25a$ and by less than 10^{-10}
 1347 at $2a$. The observers are the nine angles of Fig. 5(b) on spheres of radius
 1348 $1.25a$, $2a$ and $10a$, where k_{gr} runs from 0.6 to 10; the reference is the exact
 1349 partial-wave displacement at the same points; and the error is the measure
 1350 of the far-field test. Table 15 gives it for the mean-only voxel and for the
 1351 first-moment voxel [4]. The orders of the far field hold in the near field:
 1352 second for the mean-only voxel at every distance and both frequencies, and
 1353 3.8 to 3.95 for the first-moment voxel, still rising towards four as in the far
 1354 field at this contrast. A quarter of a radius from the surface, with 16 voxels
 1355 across, the first-moment voxel's error is 2.5×10^{-5} and 1.9×10^{-5} , 78 and
 1356 153 times below the mean-only voxel's. The error grows towards the body,
 1357 by a factor of 1.5 to 7 between $10a$ and $1.25a$, because the near field carries
 1358 the higher multipoles of the source, which a given grid resolves less well; the
 1359 order does not change. On the coarsest grid a voxel is as wide as the graded
 1360 shell and twice as wide as the gap to the nearest observer, and the sequence
 1361 is not yet asymptotic. Read at these distances with the asymptotic formula,
 1362 the same solutions would be in error by 48% to 85%: the far-field readout
 1363 is for the far field (`scripts/measure_graded_sphere_near_field.py`; the
 1364 readout is `graded_field` of the package, tested against the propagator of a
 1365 point source, against the gradient of its own displacement, and against the
 1366 far-field readout at great distance).

1367 *The voxels against an impedance march.* A second route to the same body
 1368 is invariant imbedding. The impedance of §9, Eq. (60), is marched up-

Table 15: The graded sphere read in the near field: relative error of the scattered displacement at 1.25, 2 and 10 radii from the centre, against the exact displacement at the same points, and the apparent order between the two finest grids.

n_{sub}	mean only			first moment		
	$1.25a$	$2a$	$10a$	$1.25a$	$2a$	$10a$
$k_S a = 0.5$						
4	3.3×10^{-2}	1.2×10^{-2}	5.3×10^{-3}	1.4×10^{-2}	1.6×10^{-3}	6.3×10^{-4}
6	1.4×10^{-2}	5.2×10^{-3}	1.9×10^{-3}	1.2×10^{-3}	3.8×10^{-4}	1.9×10^{-4}
8	8.0×10^{-3}	2.9×10^{-3}	1.1×10^{-3}	4.7×10^{-4}	1.4×10^{-4}	7.3×10^{-5}
12	3.5×10^{-3}	1.2×10^{-3}	4.7×10^{-4}	7.8×10^{-5}	3.1×10^{-5}	1.7×10^{-5}
16	2.0×10^{-3}	6.7×10^{-4}	2.6×10^{-4}	2.5×10^{-5}	1.0×10^{-5}	5.7×10^{-6}
order, 12 $\rightarrow$ 16	2.03	2.06	2.02	3.95	3.85	3.77
$k_S a = 1.0$						
4	4.8×10^{-2}	3.5×10^{-2}	3.3×10^{-2}	9.9×10^{-3}	1.3×10^{-3}	6.0×10^{-4}
6	2.0×10^{-2}	1.4×10^{-2}	1.4×10^{-2}	9.3×10^{-4}	3.0×10^{-4}	2.0×10^{-4}
8	1.2×10^{-2}	8.1×10^{-3}	7.8×10^{-3}	3.5×10^{-4}	1.1×10^{-4}	7.4×10^{-5}
12	5.2×10^{-3}	3.6×10^{-3}	3.5×10^{-3}	6.0×10^{-5}	2.5×10^{-5}	1.7×10^{-5}
16	2.9×10^{-3}	2.0×10^{-3}	1.9×10^{-3}	1.9×10^{-5}	8.3×10^{-6}	5.7×10^{-6}
order, 12 $\rightarrow$ 16	2.01	2.01	2.00	3.95	3.84	3.78

Table 16: The smoothly graded sphere, voxelised with the mean-only voxel on grids of n_{sub} voxels across, against its exact solution: relative far-field error, and the apparent order between successive grids. Nine unknowns per cell, every cell overlapping the sphere kept.

n_{sub}	unknowns	$k_{\text{sa}} = 0.5$		$k_{\text{sa}} = 1.0$	
		error	order	error	order
4	576	7.70×10^{-3}		3.24×10^{-2}	
6	1 872	3.04×10^{-3}	2.30	1.32×10^{-2}	2.22
8	3 672	1.68×10^{-3}	2.05	7.52×10^{-3}	1.95
12	11 520	7.30×10^{-4}	2.06	3.34×10^{-3}	2.00
16	24 984	4.06×10^{-4}	2.04	1.88×10^{-3}	2.00
20	47 304	2.58×10^{-4}	2.03	1.20×10^{-3}	2.00
24	79 344	1.79×10^{-4}	2.02	8.34×10^{-4}	2.00

wards through the depth range of the sphere with the lateral variation of the medium carried inside the march, on a laterally periodic grid, and a second sweep carries the field down [50]; the march returns the reflected plane waves on a plane above the body and the transmitted ones on a plane below. It is run here on the graded sphere and set beside the voxels, to ask how many voxels a given accuracy costs against an established alternative. Four things are fixed so that the comparison is of like with like.

1. *The planes.* Both methods are read on the planes $z = -1.25a$ and $z = +1.25a$. On the tangent planes $z = \pm a$ themselves the observers lie on the faces of the voxels at the poles, where the Gauss rule of the voxel readout (Eq. (61)) cannot integrate the singular kernel: its change from four to six nodes is 2% to 10% there, more than the scheme's error,

1381 and below 10^{-4} a quarter of a radius away. The march's amplitudes are
 1382 carried across the same homogeneous gap by each order's own phase,
 1383 which is exact.

1384 2. *The periodic array.* The march needs a laterally periodic grid, so it
 1385 solves a square array of spheres. Its answer is scored against the exact
 1386 solution of one sphere summed over the array, by Poisson summation,
 1387 which is exact but for scattering between the spheres. That part, and
 1388 the lateral resolution, are errors of the method as used on a single body,
 1389 and are counted.

1390 3. *The error.* For the march, the norm of the difference over the plane-
 1391 wave orders of the grid, relative to the norm of the exact scattered
 1392 orders; for the voxels, the norm of the difference of the scattered dis-
 1393 placement over a patch of side $5a$ on each plane, relative to the norm
 1394 of the exact one. The two are the same quantity in wavenumber and
 1395 in space, for the array and for the single body.

1396 4. *The refinement.* For the march, the lateral grid, with the depth step
 1397 refined until the error no longer changes; for the voxels, the grid and
 1398 the degree.

1399 Table 17 gives the result. The march converges quickly in the lateral grid,
 1400 because the body is smooth, and with far fewer unknowns than any voxel
 1401 scheme: at 20×20 it is at 7×10^{-4} with 1200 unknowns on a plane. The
 1402 mean-only voxel does not reach that with sixteen voxels across and twenty
 1403 times the unknowns; on a sharp sphere, where it also carries a staircase, it
 1404 fares worse. The first-moment voxel reaches the march's accuracy with six
 1405 voxels across, and below about 2×10^{-4} it continues at close to fourth order

Table 17: The graded sphere at $k_S a = 0.5$ read on the planes $z = \mp 1.25a$: relative error in reflection and in transmission. The march's unknowns are those of one plane, on which it carries a dense matrix; the period is $5a$ unless stated.

method	setting	unknowns	reflection	transmission
impedance march	12×12 , 16 steps	432	1.2×10^{-2}	9.9×10^{-3}
	16×16 , 32 steps	768	4.3×10^{-3}	3.6×10^{-3}
	20×20 , 32 steps	1200	7.6×10^{-4}	7.3×10^{-4}
	28×28 , 48 steps	2352	2.5×10^{-4}	5.6×10^{-4}
	30×30 , period $7.5a$	2700	7.2×10^{-4}	4.7×10^{-4}
mean-only voxel	8 across	3672	5.0×10^{-3}	7.9×10^{-3}
	16 across	24 984	1.2×10^{-3}	2.0×10^{-3}
first-moment voxel	6 across	7488	3.3×10^{-4}	5.2×10^{-4}
	8 across	14 688	1.4×10^{-4}	2.2×10^{-4}
	16 across	99 936	1.1×10^{-5}	1.8×10^{-5}
second-moment voxel	8 across	36 720	1.4×10^{-5}	2.3×10^{-5}
	10 across	69 120	2.6×10^{-6}	4.2×10^{-6}

1406 where the march does not: in transmission the march's error at a period
 1407 of $5a$ stops near 5×10^{-4} , which is the scattering between the spheres of
 1408 its array, since lengthening the period lowers it and refining the grid does
 1409 not. The costs are of different kinds. The march's unknowns are few but its
 1410 operator is a dense matrix on the plane, marched through the depth, and a
 1411 finer lateral grid requires a smaller depth step (with eight steps the 20×20
 1412 march diverged); the voxel schemes have many unknowns and a product done
 1413 by fast Fourier transform. In the timings of these runs, taken on one machine
 1414 with other work in progress and therefore only indicative, the first-moment
 1415 voxel at six across was solved in about a sixth of the time of the 20×20 march.
 1416 The march has what the comparison does not use: it returns the reflection
 1417 and transmission for every incident plane wave of the grid at once, and a
 1418 laterally periodic or laterally extended medium is its natural problem and
 1419 not a cost. The comparison is at one frequency (`scripts/measure_graded_`
 1420 `sphere_march.py`, `scripts/measure_graded_sphere_planes.py`).

1421 *A sphere with a sharp surface.* The homogeneous sphere is the familiar test,
 1422 and its voxelisation changes with every grid. Table 18 shows its error for
 1423 grids of 4, 6 and 8 voxels across: 4.6%, then 19%, then 4.2% at $k_S a = 0.5$.
 1424 The error tracks the staircase's volume, which is 5%, 17% and 4% too large
 1425 on these grids. Rescaled by the ratio of the sphere's volume to the staircase's,
 1426 it falls below 1% at $k_S a = 0.5$ and to at most 3% at $k_S a = 1.0$, but it still
 1427 does not converge: what remains is the staircase's shape. A refinement study
 1428 of a sharp sphere measures its staircase; the smooth sphere, like the layer,
 1429 measures the scheme.

Table 18: A homogeneous sphere voxelised on grids of n_{sub} voxels across, against exact elastic scattering: relative far-field error, raw and rescaled by the ratio of the sphere’s volume to the staircase’s. It does not converge with refinement, because the staircase changes with the grid.

n_{sub}	voxels	$k_S a = 0.5$		$k_S a = 1.0$	
		raw	volume-rescaled	raw	volume-rescaled
4	32	4.6×10^{-2}	1.4×10^{-3}	5.0×10^{-2}	5.4×10^{-3}
6	136	1.94×10^{-1}	7.9×10^{-3}	1.68×10^{-1}	3.0×10^{-2}
8	280	4.2×10^{-2}	2.3×10^{-3}	3.5×10^{-2}	8.9×10^{-3}

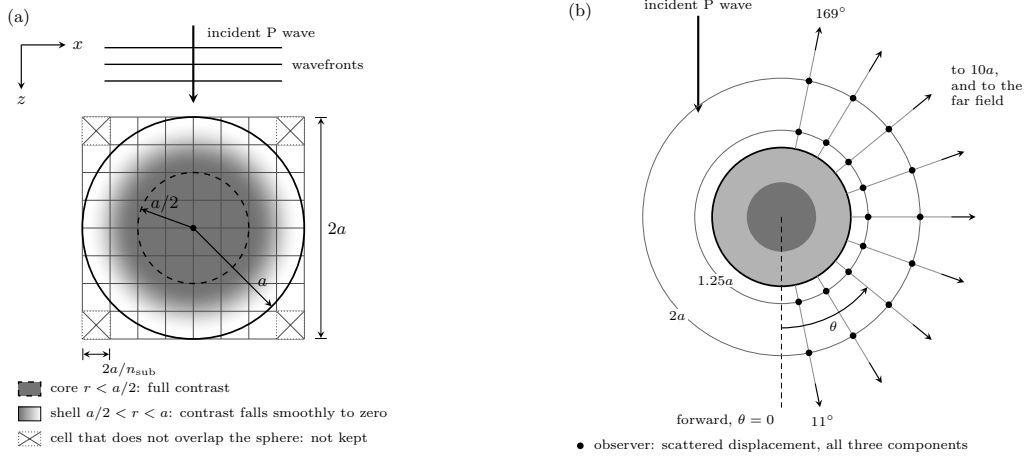


Figure 5: The sphere experiment, in the section through the centre that contains the direction of incidence. (a) The body and the grid. The contrast is full in the core $r < a/2$ and falls smoothly to zero across the shell $a/2 < r < a$. The circumscribing cube of side $2a$ is cut into n_{sub}^3 cells, here $n_{\text{sub}} = 8$; every cell that overlaps the sphere is kept and carries the projection of the contrast it contains. A P plane wave is incident along $+z$. (b) The observers. The scattered displacement is taken at nine scattering angles θ , measured from the forward direction and equally spaced from 11° to 169° , on spheres of several radii about the centre, and compared with the exact displacement at the same points.

Alt text: Two panels. On the left, a disc on a square grid of eight by eight cells: the inner half of the disc is uniformly dark, and the outer half fades from dark to white at the rim, which is drawn as a solid circle; the four corner cells of the grid, which the disc does not reach, are crossed out. Horizontal lines above the grid with a downward arrow mark the incident wave. On the right, the same disc drawn smaller, with two concentric circles around it at one and a quarter and at two radii; nine dots on each circle lie on nine rays from the disc, fanning from just off the downward, forward direction to just off the upward, backward one, and each ray ends in an outward arrow.

1430 11. The gradient hierarchy as a voxel scheme

1431 The Legendre voxels of §8 represent the field of a voxel by its moments
 1432 over the voxel. The hierarchy of §6 represents it by its gradients at the centre.
 1433 Both are polynomial, and §6.4 shows that on the layer the hierarchy of degree
 1434 $\ell = 3$ is fourth order, as the first-moment voxel is. This section builds the
 1435 hierarchy into a voxel scheme in three dimensions and compares the two.

1436 *The scheme.* Each voxel c , centred at $\mathbf{x}_c$, carries the coefficients $U_{j;W}^c$ of
 1437 Eq. (19). The equation and its derivatives to order ℓ are imposed at every
 1438 centre:

$$U_{i;P}^c - \sum_{c'} \left\{ \omega^2 \Delta \rho \sum_W \frac{K_{ij}[P;W](\mathbf{x}_c - \mathbf{x}_{c'})}{|W|!} U_{j;W}^{c'} + \sum_W \frac{K_{in}[Pk;W](\mathbf{x}_c - \mathbf{x}_{c'})}{|W|!} \delta c_{nkrj} U_{j;rW}^{c'} \right\} = \partial_P u_i^0(\mathbf{x}_c),$$

1439 (62)

$$K_{in}[D;W](\Delta) = \int_V (\partial_D G_{in})(\Delta - \xi) \xi_W d\xi.$$

(63)

1440 The two kinds of block are of different character. For $c' = c$ the integrand is
 1441 singular and the integral is a distribution: it is the moment of the hierarchy,
 1442 $K_{in}[D;W](0) = (-1)^{|D|} \mathbf{g}_{in}[D;W]$, in closed form on the cube. For $c' \neq c$
 1443 the field point is the centre of another voxel, at least half a voxel from the
 1444 source voxel, so the integrand is smooth and a tensor Gauss rule integrates
 1445 it; the derivatives of the Green's tensor, to any order, come from the family
 1446 $x^a y^b z^c r^p e^{ikr}$, which is closed under differentiation. All the singular work
 1447 of the scheme is thus in the self block, where it is done once and in closed
 1448 form; the Legendre voxel's self block is a double integral of the same singular
 1449 kernel. The blocks depend on the offset alone, so the product of the system
 1450 with a vector is a convolution over the lattice and is done exactly by the fast
 1451 Fourier transform. Where the contrast varies within a voxel it is projected

1452 there on polynomials of degree r , as in §9, and the source is the product of
1453 that polynomial and the field's: the moments then carry weights to degree
1454 $\ell + r$, and a block is a sum, over the monomials of the contrast, of blocks
1455 with the weight shifted.

1456 *Checks of the construction.* The derivatives of the Green's tensor to fourth
1457 order agree with the closed forms of its value, gradient and second derivative,
1458 and with central differences beyond them, to 8×10^{-7} . A lattice of one voxel
1459 reproduces the single site of §6.5 to 10^{-14} . Only the offsets of one octant
1460 are computed, the others following from the parity of each index; a block
1461 obtained that way equals one computed directly to 10^{-15} . The iterative
1462 solution equals a dense solution to 7×10^{-11} where both can be run.

1463 *A body with corners.* The cube of §6.5, at $k_S h = 0.2$, is cut into n^3 voxels.
1464 Table 19 gives the far-field error against the converged reference, with the
Legendre voxels on the same body. On this body every degree of the hierarchy

Table 19: A homogeneous cube cut into n^3 voxels, $k_S h = 0.2$: relative far-field error of the hierarchy of degree ℓ and of the Legendre voxels, with the unknowns of each voxel.

	unknowns a voxel	$n = 1$	2	3	4
hierarchy, $\ell = 1$	12	5.4×10^{-3}	1.7×10^{-3}	8.3×10^{-4}	5.0×10^{-4}
hierarchy, $\ell = 2$	30	1.6×10^{-3}	6.7×10^{-4}	3.7×10^{-4}	2.4×10^{-4}
hierarchy, $\ell = 3$	60	3.4×10^{-4}	7.0×10^{-5}	3.0×10^{-5}	1.7×10^{-5}
Legendre, mean only	9	5.4×10^{-3}	1.3×10^{-3}	5.8×10^{-4}	3.3×10^{-4}
Legendre, first moment	36	4.8×10^{-5}	6.0×10^{-6}		

1465

1466 converges at about the same rate, an order near two between the two finest

1467 grids, and the degree buys a constant: thirty times from $\ell = 1$ to $\ell = 3$
 1468 on 64 voxels. That rate is the body's and not the scheme's, for the mean-
 1469 only Legendre voxel shows it too: a homogeneous cube has edges and corners,
 1470 whose fields no polynomial follows. The uniform-strain closure and the mean-
 1471 only voxel agree on one cell, as they must, being the same approximation.
 1472 And the first-moment Legendre voxel is far ahead: one cell of 36 unknowns is
 1473 more accurate than eight voxels of the third-gradient hierarchy with 480. A
 1474 projection, which the Legendre voxel is, follows a field that is not polynomial
 1475 better than an expansion about a point.

1476 *A smooth medium.* The comparison that bears on voxel schemes is in a
 1477 medium without corners. First in one dimension: the layer of §6.4 with
 1478 a contrast that varies in depth, projected in each cell to the degree of the
 1479 field, against the exact graded layer. With the profile $\sin^2(\pi z/D)$ the orders
 1480 from 16 to 32 cells are 1.99, 1.98 and 3.99 for $\ell = 1, 2$ and 3: a varying
 1481 contrast leaves the pairs of §6.4 as they are. With the smoothstep of §10 as
 1482 the profile they are 1.97, 1.99 and 3.84, and on coarser grids they are erratic,
 1483 the third-gradient scheme being worse than the second on four cells. The
 1484 orders appear only when about four cells cross the graded part of the profile
 1485 (`scripts/measure_layer_taylor_hierarchy_graded.py`). That fixes the
 1486 conditions of the three-dimensional test. The graded sphere of §10 has a shell
 1487 5m wide, and a solver that reaches fourteen voxels across puts fewer than
 1488 four voxels across it; so the core is reduced here from $a/2$ to $a/10$, which
 1489 widens the shell to 9m, and a second profile, $\sin^2(\pi x/2)$ in the fraction x of
 1490 the shell's width measured from the surface, is run beside the smoothstep.
 1491 The exact reference is the radial solution of §10 for each body. The contrast

1492 is linear in each voxel for $\ell = 3$, which suffices for fourth order (§9), and of
degree ℓ for $\ell = 1$ and 2.

Table 20: The graded sphere with a shell 9m wide, $k_{Sa} = 0.5$: relative far-field error against the exact solution for n_{sub} voxels across, with the apparent order between the two finest grids of each row. The lower block is the profile $\sin^2$.

	$n_{\text{sub}} = 4$	6	8	10	12	14	order
hierarchy, $\ell = 1$	1.5×10^{-3}	8.4×10^{-4}	4.9×10^{-4}	3.2×10^{-4}			1.94
hierarchy, $\ell = 2$	3.3×10^{-4}	2.1×10^{-4}	1.5×10^{-4}	1.0×10^{-4}			1.56
hierarchy, $\ell = 3$	2.6×10^{-4}	5.0×10^{-5}	1.7×10^{-5}	7.2×10^{-6}	3.6×10^{-6}	2.0×10^{-6}	3.80
Legendre, mean only	7.3×10^{-3}	3.2×10^{-3}	1.8×10^{-3}	1.1×10^{-3}			2.02
Legendre, first moment	2.9×10^{-4}	6.4×10^{-5}	2.2×10^{-5}	9.4×10^{-6}	4.6×10^{-6}	2.5×10^{-6}	3.92
hierarchy, $\ell = 3$	1.9×10^{-4}	3.8×10^{-5}	1.3×10^{-5}	5.2×10^{-6}	2.7×10^{-6}	1.5×10^{-6}	3.84
Legendre, first moment	2.2×10^{-4}	4.8×10^{-5}	1.6×10^{-5}	6.8×10^{-6}	3.4×10^{-6}	1.8×10^{-6}	4.02

1493

1494 Table 20 gives the result. The first-gradient scheme converges at second
1495 order, and is three to four times more accurate than the mean-only Legendre
1496 voxel on every grid, for twelve unknowns a voxel against nine. The second-
1497 gradient scheme is more accurate again but gains no order, as §6.4 predicts;
1498 its apparent order is still rising towards two. The third-gradient scheme
1499 converges at an apparent order between 3.7 and 4.1 from grid to grid, 3.8
1500 over the range from six to fourteen voxels across, for both profiles; the first-
1501 moment Legendre voxel does the same, between 3.7 and 4.0 and 3.85 over
1502 the range. Neither has settled on four at fourteen voxels across, and the
1503 second profile, gentler than the first, changes neither: the two fourth-order
1504 schemes behave alike. On every grid from six voxels across the hierarchy's

error is the smaller, by a fifth to a quarter. It carries sixty unknowns a voxel against thirty-six, so that for a given number of unknowns the Legendre voxel remains somewhat the more accurate; the wide margin of Table 19 was the corners.

What the hierarchy is for. The two schemes reach the same order by different closures of the same equation, and they differ in what they cost and in what they give besides. The Legendre voxel is the more economical in unknowns. The hierarchy's blocks are simpler: its self block is known in closed form on four constants (§6.2), and its coupling is a single integral of a smooth function where the Legendre voxel's is a double integral; the relative cost of forming the two has not been measured here. The iteration converged in five steps at every size; the time went into the tables of blocks, which are computed once for a grid and a frequency. And the hierarchy is, in addition, a single-site T -matrix in closed form, whose accuracy improves block by block (§6.5), which no voxel defined by moments over the cell provides. The lattice results of this section come from one implementation, checked against exact solutions and, where it can be, against a dense solve; they are at one frequency and on one family of bodies (`scripts/gradient_voxel_lattice.py`, `scripts/measure_lattice_gradient_hierarchy.py`, `scripts/measure_graded_sphere_gradient_hierarchy.py`, `scripts/measure_graded_sphere_gradient_hierarchy_fft.py`).

12. Discussion

Figure 6 collects the convergence of every test, with the order fitted to each by least squares (`scripts/plot_convergence_orders.py`).

1529 *What each basis is for.* A voxel holds two functions, and the results give
 1530 each basis its part. The basis of the wavefield sets how well the voxel carries
 1531 the field across itself: in a uniform medium the scheme converges at order
 1532 $2p+2$ with the constant c_p of Eq. (41), for P and S waves and at normal and
 1533 oblique incidence alike. The basis of the medium sets how well the voxel *is*
 1534 the medium: with degree r the order is capped at $2r+2$ (Eq. (44)). Neither
 1535 can make up for the other, and the second-order term shows why. The field
 1536 scattered once is a copy of the medium's profile, so the wavefield basis must
 1537 carry the profile as well as the wave, and the error of that term is the relative
 1538 projection error of the profile at the lower of the two degrees (Eq. (55)). At
 1539 the frequencies of interest the wave varies little across a cell and the medium
 1540 may vary much, so it is the medium, not the wavelength, that decides the
 1541 degree and the size of the cells. Two cases bracket the choice. Where the
 1542 medium is piecewise constant on the cells, as in a layered model with voxel
 1543 faces on its interfaces, a constant is exact for the medium, the projection
 1544 error vanishes, and the degree of the field is spent entirely on the wave.
 1545 Where the medium varies smoothly within the cells, the two degrees should
 1546 be equal: a higher degree for the contrast than for the field is wasted, and a
 1547 higher degree for the field improves only the Born term. That the two degrees
 1548 can be chosen independently is the practice of higher-order electromagnetic
 1549 volume methods [1]; what the present results add is the rule for choosing
 1550 them, and the error each choice leaves.

1551 *Where the error of a voxel scheme lives.* The results separate the two ap-
 1552 proximations a voxel scheme makes. The *coupling* between voxels makes
 1553 none: averaged over the source cell it is the continuum's cell integral exactly,

1554 between planes by the form factor's zeros and within a plane by the tiling
 1555 identity, and the self term it omits is exactly the one the single site carries.
 1556 Every error therefore comes from the *single site*: from the internal field it
 1557 assumes and, where the medium varies within the cell, from the medium it
 1558 is given. Its T -matrix sums the rescattering within the voxel exactly, but
 1559 for the voxel as represented, and the split of the second-order term in §9
 1560 measures the consequence: the error sits in the part whose two scatterings
 1561 lie in one cell, and the part between cells carries at most 0.2% of it. For a
 1562 uniform field that error is the midpoint rule's, known in closed form; for a
 1563 linear field it is the next term of the same expansion. The converse is the
 1564 practical warning. A kernel that is not the cell integral (the point Green's
 1565 tensor, or a cell average that does not match the single site's closure) leaves
 1566 a local term that does not vanish as the voxels shrink (§4): the evanescent
 1567 lattice orders carry a static contribution of fixed size per voxel volume. A
 1568 scheme built that way converges to the wrong limit, and refinement cannot
 1569 reveal it, because the error is scale-invariant.

1570 *What the single site is.* The moment hierarchy of §6 turns the uniform-field
 1571 single site from a modelling choice into a derived object with a stated error.
 1572 Its static part is exact for a sphere, and for a cube it differs from the sphere's
 1573 only by the cubic split of Eq. (32). Its dynamic part departs from exact
 1574 scattering by a sphere only through the real $(ka)^2$ term of Eq. (33); the
 1575 radiation damping is exact. That term and the collocation error of Eq. (37)
 1576 have the same origin, the variation of the field across the scatterer, and the
 1577 same remedy, a richer internal field. In a lattice the remedy is cheap: the
 1578 field a voxel must represent is the smooth continuum field, not the field of

1579 an isolated cube, with its concentrations at the edges and corners.
 1580 *Relation to the coupled-dipole method.* The same partition appears in the
 1581 coupled-dipole method as the choice of polarisability. The Clausius–Mossotti
 1582 relation, the lattice dispersion relation [11], the polarisability taken from the
 1583 exact static solution [12] and the integrated Green tensor [13] all adjust
 1584 the single site so that the lattice reproduces the continuum. For space-
 1585 filling cubes the adjustment is neither needed nor a free parameter: the tiling
 1586 identity fixes the local field exactly, the self term of the collocation closure is
 1587 the one that makes it cancel, and what is left is a closed-form error rather than
 1588 an empirical one. That the method is a collocation of the volume integral
 1589 equation was shown for the electromagnetic case by Lakhtakia [9]; Eq. (36)
 1590 is the same statement for the elastic lattice with the self term evaluated over
 1591 the finite cube, and it is what makes the error a midpoint-rule error. The
 1592 filtered coupled dipoles of Piller and Martin [14] and Yurkin et al. [15] band-
 1593 limit the source to the grid; the source-cell average does the complementary
 1594 thing exactly, its form factor vanishing on the reciprocal lattice, and the
 1595 spectral construction of Eq. (42) is the periodic coupled-dipole method of
 1596 Chaumet et al. [16] with the Galerkin double average in place of the point
 1597 dipole. The second-order convergence reported for the DDA by Yurkin et al.
 1598 [3] corresponds to the uniform-field voxel’s, and the opposite-signed constants
 1599 of Eq. (37) explain why a polarisability correction can improve one observable
 1600 only at the expense of another, as the package’s own form-factor corrections
 1601 do in reflection and transmission. Going beyond second order requires a richer
 1602 internal field, which is the point of higher-order bases for volume integral
 1603 equations [52, 2]; the first-moment voxel is the lowest such basis, and in its

1604 spectral form the Galerkin double average makes the lattice sum converge
1605 without Ewald splitting.

1606 *Relation to the spectral element method.* The voxel of degree p is close kin to
1607 the spectral element [53], which was taken up for seismic wave propagation
1608 in the 1990s, is set out for three-dimensional models by Komatitsch and
1609 Tromp [54], and is now a standard tool for it. Both divide the medium
1610 into hexahedra, represent the field in each by polynomials of high degree,
1611 and determine the coefficients by a Galerkin condition on a weak form. The
1612 spectral element uses Lagrange interpolants through the Gauss–Lobatto–
1613 Legendre points of the element; the voxel uses the Legendre polynomials
1614 themselves. Four of the present results have their counterparts there.

1615 1. *A diagonal mass matrix.* The spectral element obtains one by integrat-
1616 ing at the Gauss–Lobatto–Legendre points, at which each interpolant
1617 vanishes but one [54]. That rule is not exact for the product of two
1618 interpolants, so that the diagonal form is reached, in their words, by a
1619 process of subintegration, as in the finite-element methods that lump
1620 the mass. Here the Gram matrix $\mathbf{M}$ of Eq. (52) is diagonal with no
1621 approximation, because the Legendre functions are orthogonal on the
1622 cell; this is why the first-order term needs no solve. The two routes
1623 serve different ends: theirs an explicit march in time, ours the terms of
1624 the scattering series.

1625 2. *The doubled order.* A scattered amplitude is a functional of the field,
1626 and a Galerkin method gives a functional to twice the order of the field
1627 itself. The order $2p + 2$ is this doubling: the leading error of Eq. (40) is
1628 the product of the part of the incident wave that the cell’s polynomials

miss and the part of the outgoing wave that they miss, each of order h^{p+1} .

3. *The medium to the degree of the field.* In a spectral element the density and the elastic moduli are evaluated at the element's own grid points and may vary from one to the next, so that a fully heterogeneous medium is held to the degree of the field [54]. The rule $\min(2p+2, 2r+2)$ says why that is the right choice, and what a coarser representation of the medium would cost.

4. *Elements that honour the interfaces.* A spectral-element mesh is designed to honour the free surface and the main geological boundaries of the model [54]. The stratified layer of §9 is the same requirement seen from the integral side: with voxel faces on the interfaces the projection error vanishes, and a jump inside a voxel costs the order.

The two methods differ as a differential formulation differs from an integral one, and they suit different problems. The spectral element discretises the weak form of the equations of motion in the time domain. Its coupling is local and its matrices are sparse; it treats heterogeneity, anisotropy and topography of any form throughout a three-dimensional model; and the price is that the whole model is meshed, that the outgoing waves are absorbed at an artificial boundary, and that over a long simulation the accuracy is governed by the time scheme [54]. The voxel scheme discretises the volume integral equation in the frequency domain. Only the departure from a stratified background is meshed, the field of each voxel is independent of its neighbours', and propagation through the background, the radiation condition included, is carried exactly by the background's Green's tensor; the price is a cou-

1654 pling between every pair of voxels, and a background restricted to one whose
 1655 Green’s tensor is known. It differs also from the boundary integral methods,
 1656 which are restricted to a finite number of homogeneous regions [54]: a vol-
 1657 ume integral equation admits a medium that varies within the scatterer. The
 1658 voxel scheme is therefore a candidate where the heterogeneity is localised in
 1659 a layered medium, and not a replacement for the spectral element in a model
 1660 that is heterogeneous throughout. A spectral-element mesh is designed by a
 1661 rule of resolution, about five grid points to the shortest wavelength at poly-
 1662 nomial degrees between five and ten [54]. The error analysis of this paper is
 1663 the counterpart of that rule for the integral formulation: it gives the size and
 1664 the degree of the voxels for a required accuracy in closed form, and it adds a
 1665 term, the projection error of the medium.

1666 *The tension between the two error sources.* The coupled-dipole method has
 1667 two sources of error, the single site and the coupling between sites, and each
 1668 has been corrected on its own: the polarisability by the Clausius–Mossotti
 1669 relation, the lattice dispersion relation or the exact static limit, and the
 1670 coupling by integrating or filtering the Green tensor. The two are not in-
 1671 dependent. The point kernel’s evanescent orders carry a static local term
 1672 whose size per voxel volume does not shrink with the voxels (§4), and the
 1673 single site must cancel exactly that term. A polarisability tuned for one
 1674 kernel is therefore wrong for another: the lattice dispersion relation corrects
 1675 the point-dipole lattice, and changing the kernel to an integrated or a fil-
 1676 tered one changes the term it must cancel. Tuning either alone moves the
 1677 error between the two, and refinement cannot reveal a mismatch, because
 1678 the surviving term is scale-invariant. For space-filling cubes the tension is

1679 resolved rather than balanced: the cell-averaged kernel omits exactly the self
 1680 term that the closure carries (Eq. (36)), so the coupling error vanishes iden-
 1681 tically and every remaining error is the single site's, the variation of the field
 1682 across the cell. That error has a second tension of its own. It is a midpoint-
 1683 rule error whose constants have opposite signs in reflection and transmission
 1684 (Eq. (37)), so any scalar adjustment of the single site that improves one wors-
 1685 ens the other; the package's far-field corrections reduce the reflection error
 1686 by a factor 4.45 and increase the transmission error by 1.7. No polarisability,
 1687 however chosen, can remove it. What removes it is a richer internal field:
 1688 the first moment lowers the error by two orders in kd in both observables at
 1689 once.

1690 *The cost of fourth order.* A first-moment voxel carries four times as many
 1691 unknowns as a uniform one (36 against 9 in three dimensions). The closed
 1692 forms show what that buys. To reach an error ε in reflection, the uniform
 1693 voxel needs $kd = \sqrt{8\varepsilon}$ and the first-moment voxel $kd = (720\varepsilon)^{1/4}$; at $\varepsilon = 10^{-6}$
 1694 that is $kd = 2.8 \times 10^{-3}$ against 0.16, voxels 58 times larger in each direction.
 1695 In a three-dimensional model the voxel count falls by $58^3 \approx 2 \times 10^5$ and the
 1696 number of unknowns by about 5×10^4 (these are the Born-order constants;
 1697 at the contrast studied the first-moment voxel's measured errors lie within
 1698 5% of them, while the uniform voxel's transmission constant grows with the
 1699 contrast, so the comparison is if anything conservative). The stratified model
 1700 of §9 measures it directly: 347 planes of uniform-field voxels against 8 first-
 1701 moment voxels for the same accuracy. Higher moments continue the trend
 1702 in one dimension: by Eq. (41) the second-moment voxel reaches $\varepsilon = 10^{-6}$ at
 1703 $kd = 0.68$ and the third-moment voxel at $kd = 1.5$, where a voxel is already

1704 a quarter of a wavelength. In three dimensions, with the Legendre products
 1705 of total degree at most p , a voxel carries 10 ($p = 2$) or 20 ($p = 3$) weights per
 1706 state component, against 4 for the first-moment voxel. The voxel of degree 2
 1707 converges at sixth order in the layer at oblique incidence (Table 10); that of
 1708 degree 3 has not been built in three dimensions. The gain is largest precisely
 1709 where it matters: when the model varies on scales that are coarse compared
 1710 with the accuracy the wavefield must have.

1711 *What the error analysis offers a fast solver.* A voxel scheme is dense, and
 1712 for a large medium it is solved iteratively with a fast product. The adaptive
 1713 integral method [55] and the precorrected fast Fourier transform [56] are
 1714 the standard forms: the sources are projected onto a regular grid, in the
 1715 first by matching their multipole moments and in the second by matching
 1716 the potentials they produce at a set of test points, the interactions between
 1717 distant cells are evaluated on that grid by fast Fourier transforms, and those
 1718 between near cells are computed directly and corrected. Such a method has
 1719 three accuracies to set, that of the discretisation, that of the projection, and
 1720 the radius of the near zone, and they are usually set by trial. The results of
 1721 this paper set them beforehand, in five ways.

- 1722 1. *The target is known.* The discretisation error is known before the solve,
 1723 in order and in size: the first-order factor of a cell, Eq. (40) and its
 1724 form for any pair of wavenumbers (§9), and the projection error of the
 1725 medium, Eq. (55). A fast product need be no more accurate than that,
 1726 and it is known when it is accurate enough.
- 1727 2. *The two errors do not mix.* The error of the scheme is the single site's:
 1728 the coupling between cells, averaged over the cells, is exact (§5). An

1729 approximation of the far coupling to a set tolerance therefore adds its
1730 own error to the discretisation's and does not alter it, so the two can
1731 be budgeted separately.

1732 3. *For cubes on a lattice there is nothing to project.* The cells are the reg-
1733 ular grid. The coupling depends on the separation of the cells alone, so
1734 the product is a convolution, block by block in the polynomial degrees,
1735 and the fast Fourier transform applies to it exactly, as in Georgakis
1736 et al. [2] for the linear voxel. What the projection onto the grid ap-
1737 proximates in general is here an identity.

1738 4. *For cells of several sizes the moments are already in hand.* Where the
1739 cells differ in size or degree and are mapped to a uniform auxiliary
1740 grid, the quantities to be matched are the polynomial moments of a
1741 cell, which are the unknowns of the scheme; and the first-order factor
1742 says how many of them matter at a given kd .

1743 5. *A stratified background splits as the method requires.* The Green's ten-
1744 sor of a plane-stratified background is the whole-space tensor, which is
1745 invariant under translation in three dimensions and carries all the sin-
1746 gularity, plus a reverberation, Eq. (58), which is smooth, is invariant
1747 under lateral translation, and separates in depth into one-dimensional
1748 moments. The first is the convolution above; the second is a fast Fourier
1749 transform laterally and a product in depth, with no near-zone correc-
1750 tion except where a voxel touches an interface. A splitting of the kernel
1751 into a term that carries the singularity and a nonsingular remainder is
1752 used to a related end by Sharma and Triverio [57], for surface integral
1753 equations in homogeneous and in layered backgrounds: there the sin-

1754 gular term is independent of frequency, so that the direct near-zone
 1755 integrals are computed once for a whole sweep in frequency. And be-
 1756 cause the orders and the error law are unchanged by the background
 1757 (§9), the error budget of the whole space carries over to it.

1758 No such solver is built here. These are consequences of the analysis, and they
 1759 are to be tested.

1760 *Limitations and outlook.* The results concern laterally uniform media, a ho-
 1761 mogeneous layer and a randomly stratified one, for which the discrete system
 1762 is exactly a chain of planes; a laterally heterogeneous medium needs the first-
 1763 moment voxel in a full three-dimensional solver, where the gradient channels
 1764 couple through the evanescent lattice orders (§8) and the tiling identity holds
 1765 only to $O(k_{\parallel}h)$. The smoothly graded sphere of §10 shows the mean-only
 1766 voxel converging at second order on a laterally heterogeneous body against
 1767 an exact solution, and the companion paper [4] shows the first-moment voxel
 1768 converging at fourth order there. A contrast that varies within a cell, which
 1769 §9 shows is needed beyond second order in a smoothly varying medium, has
 1770 a further consequence in three dimensions: its linear part is odd about the
 1771 cell's centre, so it couples the single site's uniform-field and first-moment
 1772 blocks, which parity keeps apart for a homogeneous cube (§6.3). A graded
 1773 voxel therefore needs at least the first-moment block; that voxel is built,
 1774 and tested on the graded sphere, in the companion paper. The numerical
 1775 evidence is at moderate, physically typical contrasts (up to 15% in density
 1776 and moduli) and at $k_S h \leq 0.3$, the range in which the single-site T -matrix
 1777 is validated; the closed forms are first order in the contrast, and the depar-
 1778 tures from them at the full contrast are of second order. What is established

1779 is the *order* of convergence and its leading constant, not the behaviour at
 1780 strong contrast, where the single site's own closure may limit the accuracy
 1781 before the lattice coupling does. The single site's dynamic error is in closed
 1782 form for a sphere only; for a cube, for which no exact solution exists, it is
 1783 known through the layer (§§7–8) and, for the isolated cube, by comparison
 1784 with a converged solution (§6.5). The hierarchy beyond the first gradient is
 1785 established less widely than the rest: its orders and its error factor on the
 1786 layer are derived symbolically and checked independently to the third degree,
 1787 and its moments are confirmed by two routes, but its three-dimensional re-
 1788 sults, the single site against the sphere and the cube and the voxel scheme of
 1789 §11, come from one implementation, at one frequency for the voxel scheme,
 1790 checked against exact solutions. There the fourth order is approached and
 1791 not attained on the grids reached (3.8 at fourteen voxels across, as for the
 1792 first-moment voxel), the fifth-gradient system, whose moments are in hand,
 1793 has not been assembled in three dimensions, and the cost of forming the
 1794 blocks of the two fourth-order schemes has not been compared. Two exten-
 1795 sions follow directly. The first is the three-dimensional solver itself. The
 1796 second is a stratified *background*, a layered reference medium, as distinct
 1797 from the stratified contrast of §9. For a laterally uniform layer of voxels the
 1798 orders, the first-order factor and the projection-error law all survive it, at
 1799 normal and at oblique incidence, for P, SV and SH waves, in a background
 1800 of two half-spaces and in one of three layers (§9). What remains is a body
 1801 of finite lateral extent in such a background, where the image of a voxel in
 1802 an interface it touches is singular.

1803 13. Conclusions

1804 A voxel of a volume-integral scheme must represent the medium within it
 1805 and the wavefield within it. This paper has determined what each basis must
 1806 be and what error a given choice leaves, by an experiment in which nothing
 1807 else is approximated: a layer of cubes is the layer itself, so its exact solution
 1808 isolates the discretisation error.

1809 With the wavefield expanded to Legendre degree p and the contrast of the
 1810 medium to degree r in each voxel, the scheme converges at order $\min(2p +$
 1811 $2, 2r + 2)$. In a uniform layer the leading error is $c_p(kd)^{2p+2}$ with the closed-
 1812 form constant of Eq. (41), and a single plane of such voxels reproduces the
 1813 thin-layer series term by term through $O(D^{2p+2})$; the orders 2, 4 and 6 of
 1814 the degrees 0, 1 and 2 hold for P, SV and SH waves at normal and oblique
 1815 incidence. The first-moment voxel's leading error, $-(kd)^4/720$, does not
 1816 depend on the direction of observation, whereas the collocation scheme's is
 1817 opposite in sign in reflection and transmission, so that no scalar correction of
 1818 the single site removes it. Where the medium varies within a cell, a contrast
 1819 held constant caps every voxel at second order, the coupling for a contrast
 1820 of degree r needs nothing beyond the same kernel moments to degree $p + r$,
 1821 and the error of the term of second order in the contrast is the relative
 1822 projection error of the medium at degree $\min(p, r)$, exactly in the long-wave
 1823 limit. That error is the single site's: its T -matrix sums the rescattering
 1824 within a voxel exactly for the voxel as represented, and the represented voxel
 1825 is not the medium. The two bases must therefore be matched. Where the
 1826 medium is piecewise constant on the cells the projection error vanishes: the
 1827 orders survive the interfaces of a randomly stratified model, provided voxel

1828 boundaries coincide with them, and there the first-moment voxel reaches an
1829 accuracy of 10^{-6} with one voxel per model layer, where the uniform-field
1830 voxel needs about forty. A sphere whose contrast falls smoothly to zero at
1831 its surface converges at second order against its exact solution, as the layer
1832 predicts, to 0.018% at $k_{sa} = 0.5$ with twenty-four voxels across; a sharp
1833 sphere's error is set by its staircase, which changes with the grid.

1834 Three properties of the cube make the experiment exact. Because cubes
1835 tile the plane, the source-cell-averaged coupling between voxels is the con-
1836 tinuum's cell integral exactly, and at normal incidence the lattice kernel is
1837 available in closed form with no lattice sum. The single site is derived rather
1838 than assumed: the moments of the Green's tensor over the cube, taken as
1839 distributions, give a closed hierarchy whose first-gradient truncation is the
1840 uniform-strain closure. That closure is exact for a sphere in the static limit,
1841 it departs from exact scattering by a sphere only by a real $(ka)^2$ term, with
1842 the radiation damping exact, and for a cube it differs from the sphere only
1843 by a cubic anisotropy of the shear response. And the collocation closure
1844 carries the same self term as the lattice kernel, which cancels, so the discrete
1845 system is a collocation of the continuum equation, whose second-order error
1846 is a midpoint-rule error of the internal field, known in closed form for every
1847 discretisation.

1848 The uniform-strain closure is the first member of a hierarchy. Expanding
1849 the field of a voxel in its gradients at the centre, to any degree, gives a
1850 closed system whose coefficients are moments of the Green's tensor over the
1851 cube, graded by the number of derivatives and the degree of the weight; every
1852 moment through the fifth degree is in closed form on the four constants $1, \sqrt{3},$

1853 π and $\log(2 - \sqrt{3})$, gated by sum rules that test its distributional content and
 1854 confirmed by an independent evaluation. The blocks of the hierarchy gain
 1855 accuracy in pairs: the second gradients complete the density channel and the
 1856 third the stiffness channel, so that fourth order needs the third gradients and
 1857 sixth the fifth, and the error of each truncation at first order in the contrast is
 1858 in closed form, the constants of the collocation scheme being its first case. As
 1859 a single site the third-gradient system is three hundred times more accurate
 1860 than the uniform-strain closure against the exact sphere at $k_S a = 0.4$, and
 1861 lowers the static error of an isolated cube ninefold. As a voxel scheme, with its
 1862 self block in closed form and its coupling a convolution of smooth integrals, it
 1863 converges on a smoothly graded sphere as the first-moment Legendre voxel
 1864 does, and is the more accurate of the two on a given grid by a fifth to a
 1865 quarter, for sixty unknowns a voxel against thirty-six. The results, and the
 1866 evidence for each, are collected in Table 21.

1867 *The error is known in advance.* For every discretisation studied here the
 1868 error is not something to be discovered by refining until the answer stops
 1869 changing. Its order and its leading constant are known before anything is
 1870 computed: $+(kd)^2/8$ in reflection and $-(kd)^2/24$ in transmission for the
 1871 collocation voxel (Eq. (37)), $(kd)^2/12$, $(kd)^4/720$ and $(kd)^6/100800$ for the
 1872 voxels of degree 0, 1 and 2 (Eq. (41)), and the order $\min(2p + 2, 2r + 2)$
 1873 where the medium varies within a cell (Eq. (44)), with the size of the error
 1874 there given by the relative projection error of the medium (Eq. (55)), which
 1875 is computed from the model and the grid alone. That is worth more than
 1876 the order alone, for three reasons. First, the cell size and the degree of the
 1877 voxel that a required accuracy needs follow from the wavenumber and the

Table 21: The results of the paper, and the evidence for each.

Result	Evidence
Continuous layer, exact and to $O(D^k)$	reference to 2.5×10^{-8} ; series order k
Thin-layer series, Eq. (7)	each coefficient symbolically, to D^4
Point-kernel local term survives $d \rightarrow 0$	$11 \times$ the continuum term at $m = 1$
Averaged coupling between planes = continuum $\times$ sinc	2.8×10^{-16} , 81 entries
Tiling identity, Eq. (13)	21/21 components to 4×10^{-9}
Exact kernel, Eq. (14)	replaces a 9.4×10^{-4} static error
Moments as distributions, Eq. (25)	sum rule (26) exact; M, Q equal an independent archive, 81/81
Channels (29)–(31)	a production code by another route, 2×10^{-15}
Cube vs sphere, Eq. (32)	symbolic; and numerically at five ν , 4×10^{-16}
Ball closure = exact Mie, static	2×10^{-10} ; cube’s bulk and shear average 10^{-8}
Ball closure vs Mie, dynamic, Eq. (33)	symbolic; numerical Mie 5×10^{-4} ; $(ka)^3$ terms exact
Self term cancels, Eq. (36)	4×10^{-16}
Package T = closure statically	2×10^{-5} at $k_S h = 0.01$
Second-order convergence	order 2.000, three frequencies
Closed-form error, Eq. (37)	$n = 1$ –16, to 10^{-4} of the error
First-moment voxel, Eq. (39)	order 4.00, three frequencies
First-moment voxel, oblique, Eq. (42)	order 4, P and S, 20° and 40°

1878 medium, before the computation and without a refinement study. Secondly,
1879 it turns a refinement study from a hope into a test. A scheme of this kind can
1880 converge smoothly to the wrong limit (§4), and smooth convergence alone
1881 cannot reveal it; but the difference between two grids must have the known
1882 order and the known constant, and a computation whose differences do not
1883 is in error from some cause other than the discretisation. Thirdly, it shows
1884 where effort pays. The error is that of representing the field and the medium
1885 within a cell, so raising the degree of the cell does what multiplying the cells
1886 cannot: in the stratified model one first-moment voxel per layer reaches 10^{-6}
1887 where the uniform-field voxel needs about forty. Whether the error of a body
1888 with three-dimensional structure can be known in advance in the same way
1889 is the question taken up in the companion paper [4].

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1892 the Research School of Earth Sciences of the Australian National University,
1893 Canberra [21].

1894 **Declaration of competing interest**

1895 The author declares no known competing financial interests or personal
1896 relationships that could have appeared to influence the work reported in this
1897 paper.

1898 **Declaration of generative AI and AI-assisted technologies in the**
1899 **manuscript preparation process**

1900 During the preparation of this work, the author used Claude (Anthropic)
1901 in order to check the existing Mathematica notebooks, Python and Fortran
1902 code, to build tests to ensure their correctness, and to write the TikZ code
1903 of the diagrams. After using this tool, the author reviewed and edited the
1904 content as needed and takes full responsibility for the content of the published
1905 article.

1906 **Data availability**

1907 The Mathematica notebooks and Python scripts that derive and check
1908 every result, and the package they test, are available at [https://github.](https://github.com/tmnestor/MultipleScatteringCalculations)
1909 [com/tmnestor/MultipleScatteringCalculations](https://github.com/tmnestor/MultipleScatteringCalculations).

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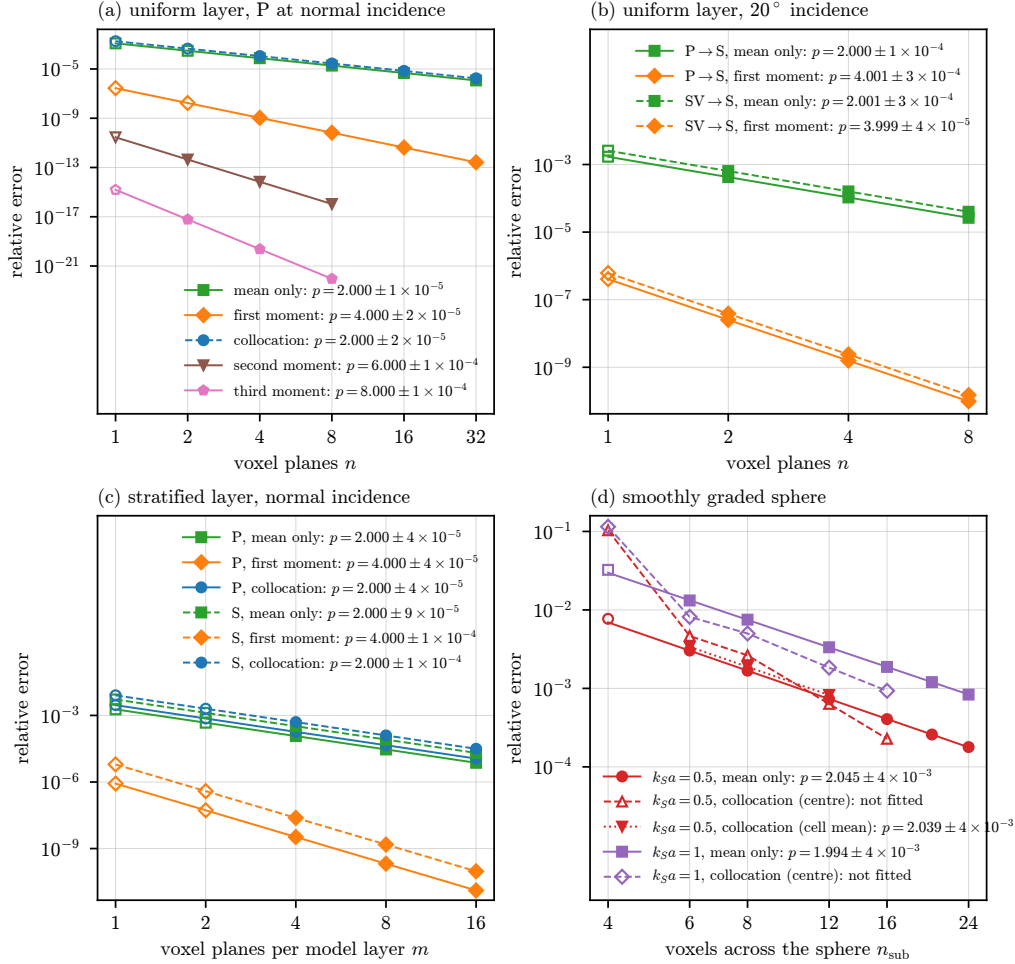


Figure 6: Convergence of every test, with least-squares regression lines. Each line fits $\log e = c - p \log n$ to the filled points; open points are pre-asymptotic and are not fitted. The legend gives the order p and its standard error. The marker, and in colour the hue, marks the scheme, so that the figure reads in grey scale: collocation (circles; §6), the mean-only voxel (squares) and the voxels carrying Legendre moments to degree $p = 1$ (diamonds), 2 (triangles) and 3 (pentagons) (Eqs. (39) and (42)). (a) The uniform layer, P reflection at normal incidence ($D = 2\text{ m}$, $\omega = 300\text{ rad s}^{-1}$, the contrast of §7), fitted from $n = 4$; the collocation line is dashed and lies on the mean-only line, from which its constant differs only by the factor $3/2$ of Table 7; the second- and third-moment voxels, computed in 40-digit arithmetic, fitted from $n = 2$. (b) The uniform layer at 20° , fitted from $n = 2$: the S wave converted from incident P (solid) and reflected from incident SV (dashed). (c) The stratified layer of §9, reflection at normal incidence for P (solid) and S (dashed), against the number m of voxel planes per model layer, fitted from $m = 4$. (d) The smoothly graded sphere of §10 against its exact solution, for n_{sub} voxels across: the mean-only voxel, fitted from $n_{\text{sub}} = 6$ (at 4 a voxel is as wide as the graded shell); the collocation voxel with the contrast sampled at the cell centre, joined by dashed lines and